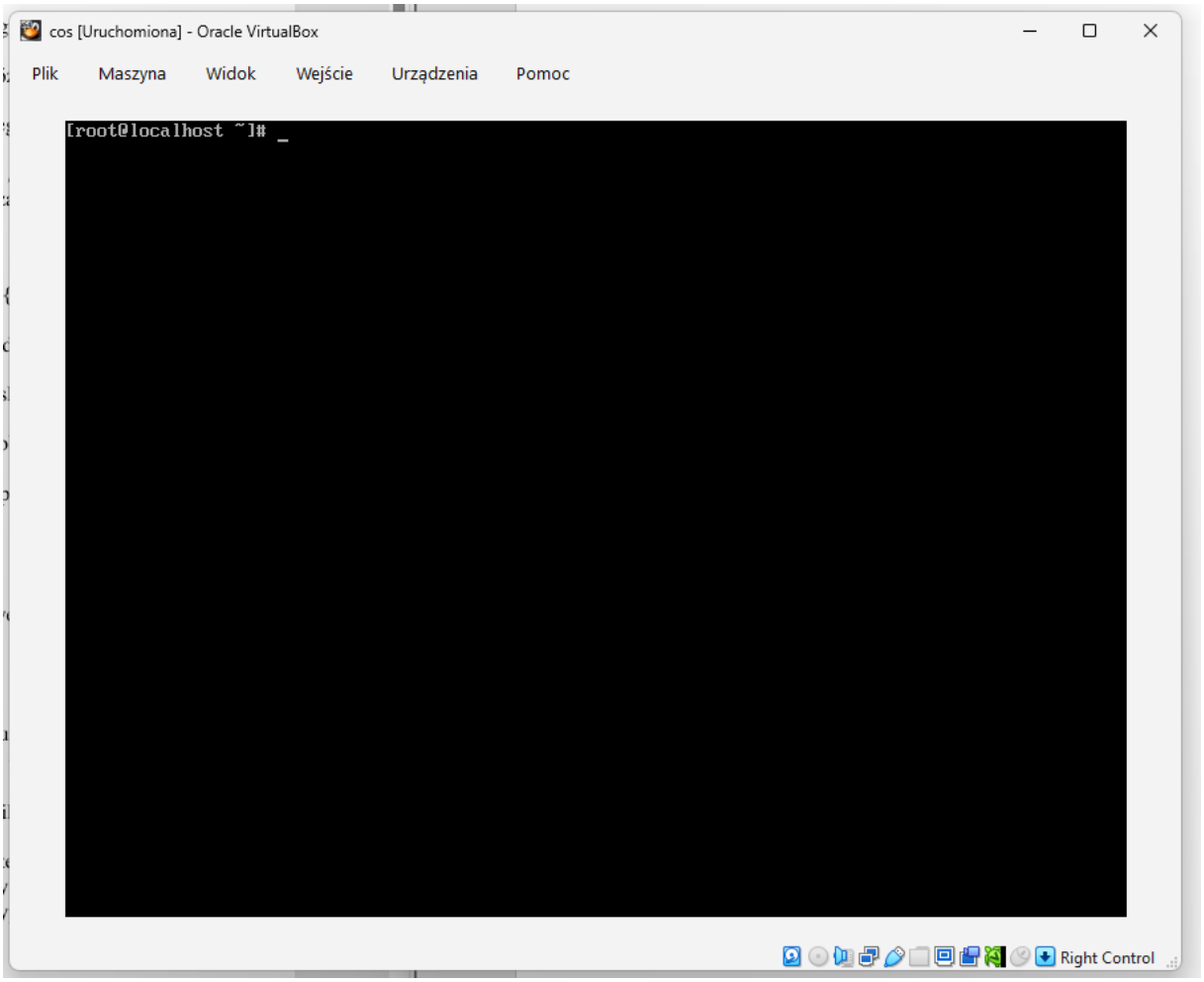
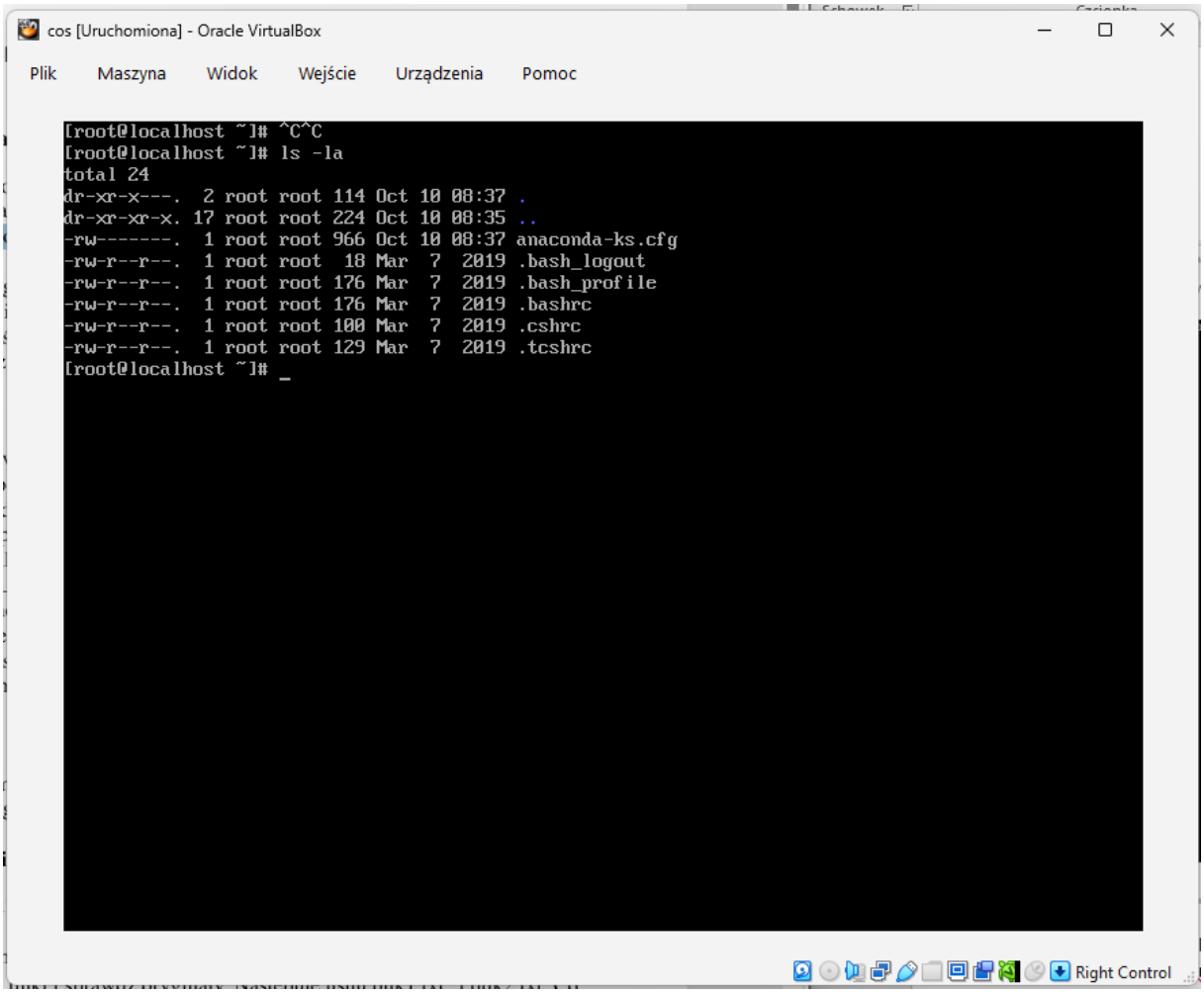
Tomasz Ceremuga

Lista 1

1. 

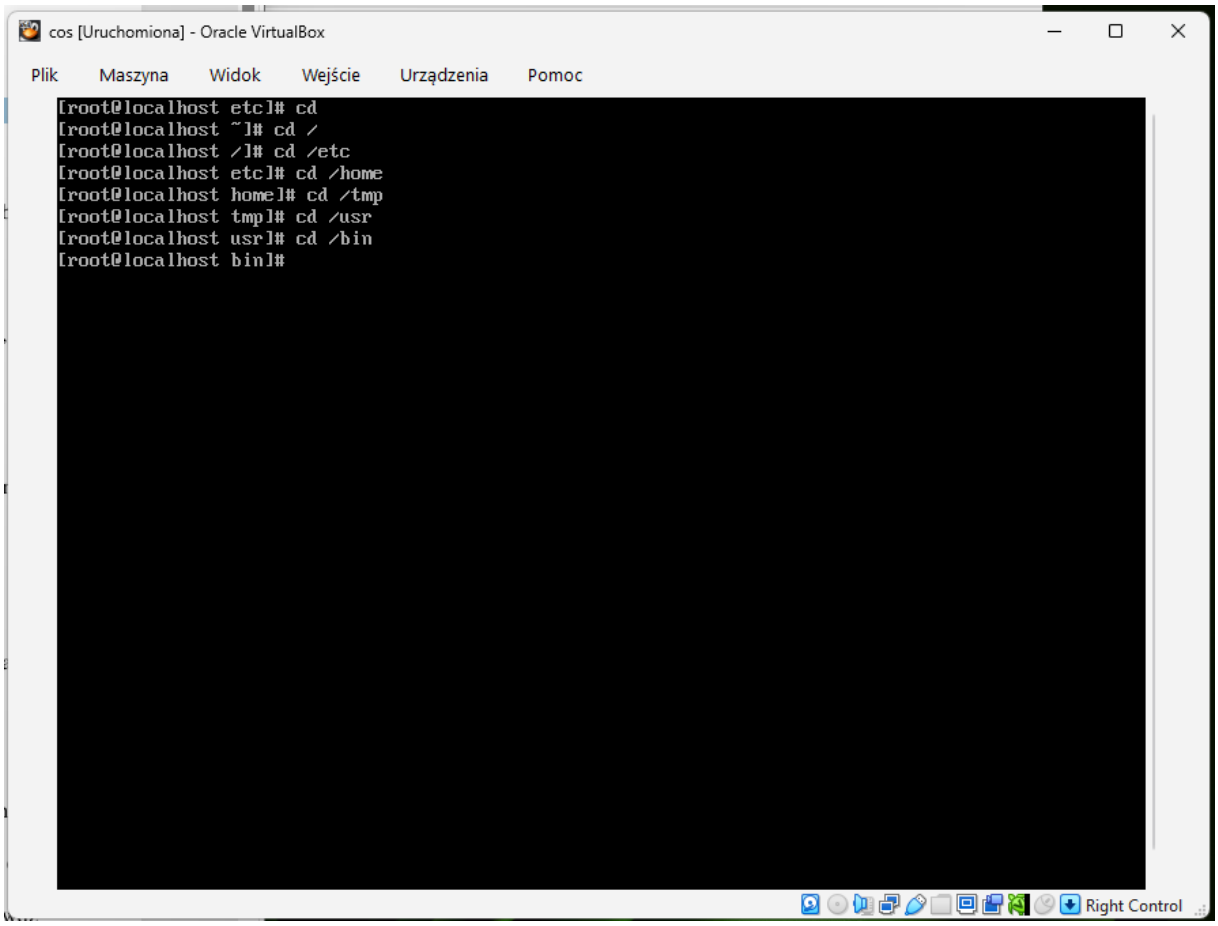
   Jesteśmy w katalogu home ~
   użytkownik root

2. 

ls – wypisuje zawartość
-l - znaczy że wyświetlamy w formacie długim, czyli szczegółowe informacje o plikach
-a - all czyli uwzględniamy ukryte pliki

Czyli ls -la wylistuje całą zawartość w formie długiej

3.



/ - root, tutaj mamy zawartość całego systemu, czyli wszystkie kolejne katalogi się w nim znajdują

/etc – tutaj znajduje się konfiguracja samego linuxa czyli konfiguracje sieci, użytkowników

/home – tutaj znajdują się katalogi domowe wszystkich użytkowników (~ to jest katalog domowy tego kto obecnie używa komputera)

/tmp – pliki tymczasowe, usuwane po reboocie

/usr – tutaj miejsce na dodatkowe programy, biblioteki, czcionki itd.

/bin – pliki binarne, pliki wykonywalne, stąd komputer wie co ma odpalić \

4. Ścieżka bezwzględna (/) – nie ważne w jakim katalogu wpisywane jest polecenie, początkiem jest root

   Ścieżka względna (.) – tutaj odwołujemy się nie od home tylko danej ścieżki w jakiej obecnie jesteśmy

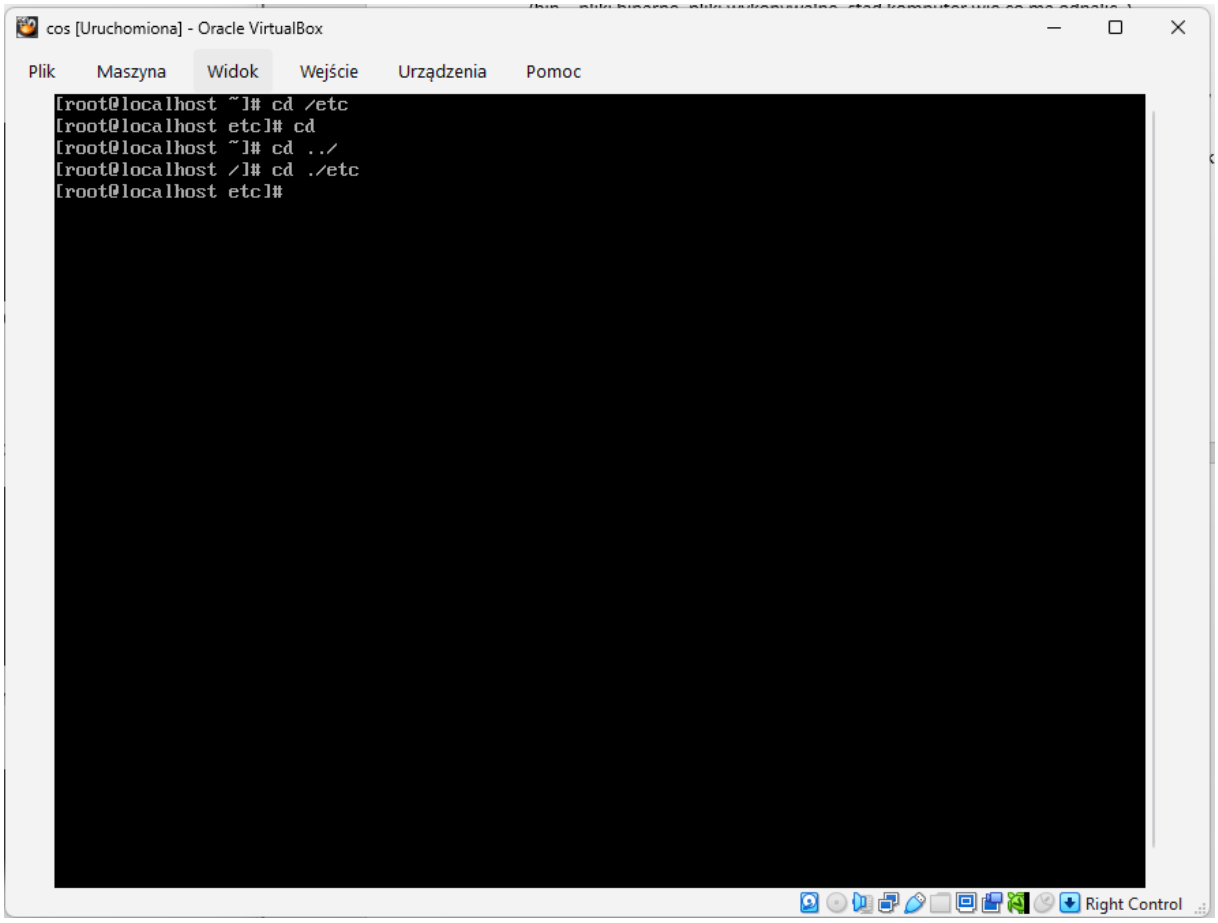
   .. – to oznacza jakby jeden katalog wyżej niż obecnie

Jeżeli steśmy w

~/dir1/dir2

~ - zawsze aliast do własnego katalogu home
. – obecny katalog czyli dir2

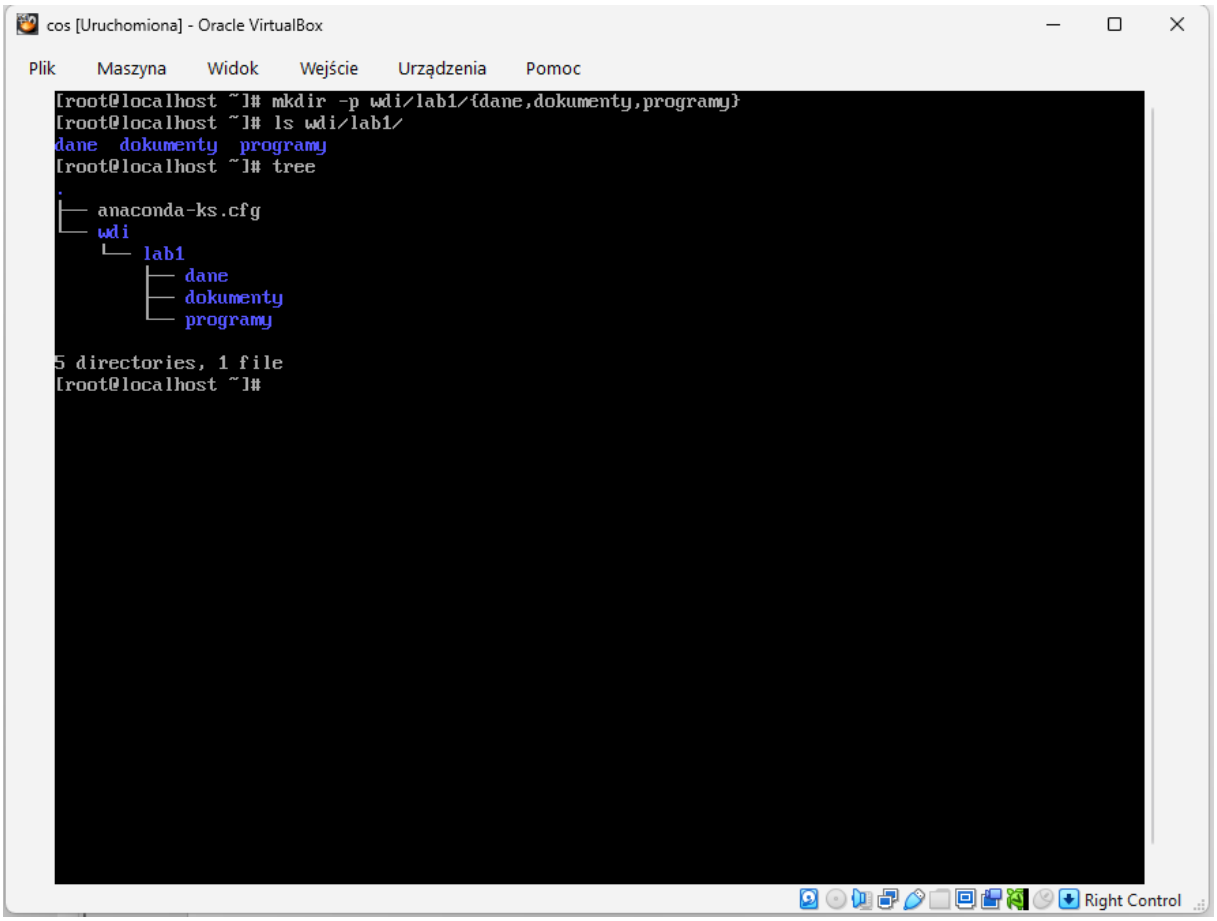
.. – katalog wyżej czyli dir1



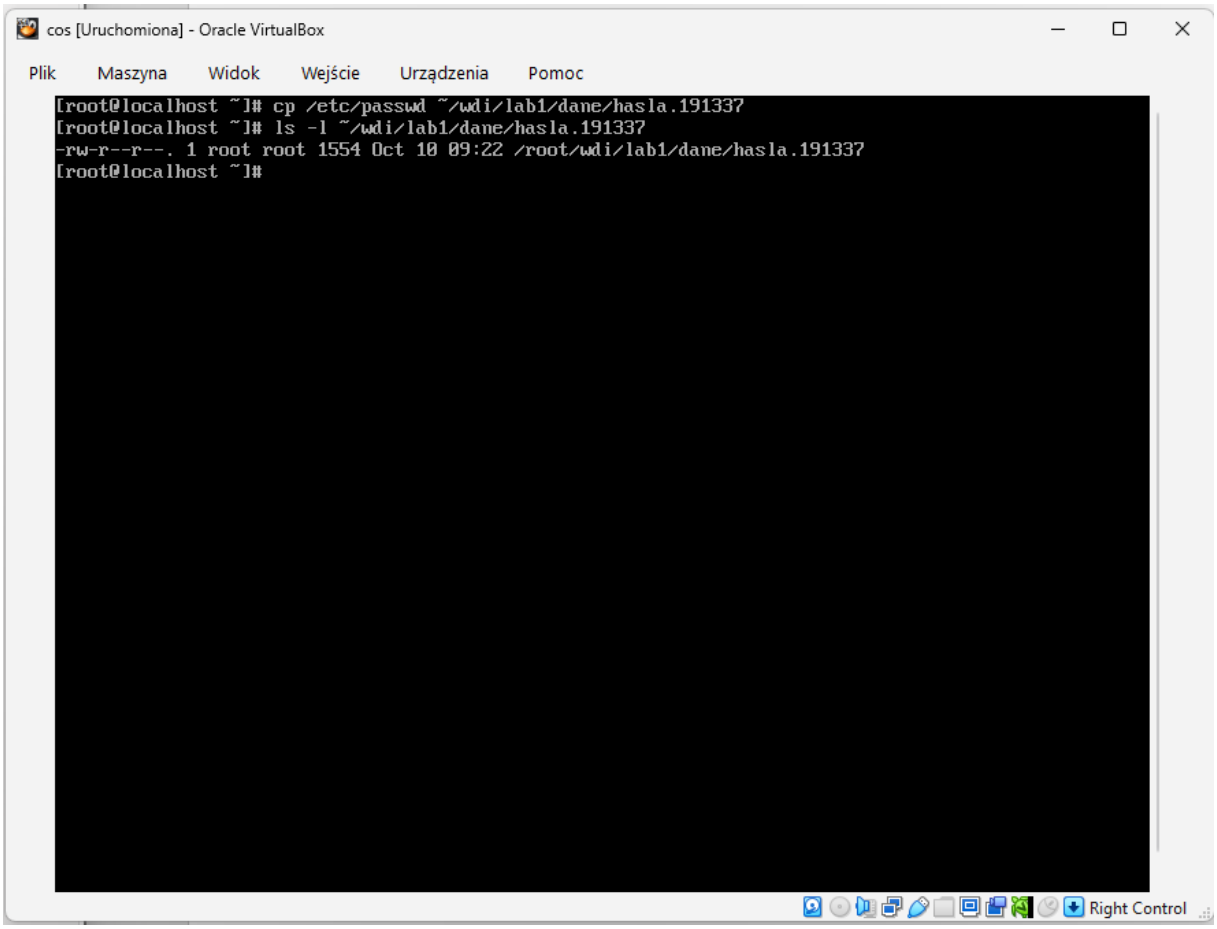
W pierwszym sposobie poszliśmy bezpośrednio od roota do etc
w drugim musieliśmy się cofnąć z home do / następnie mogliśmy przejść do podścieżki etc

Część 2

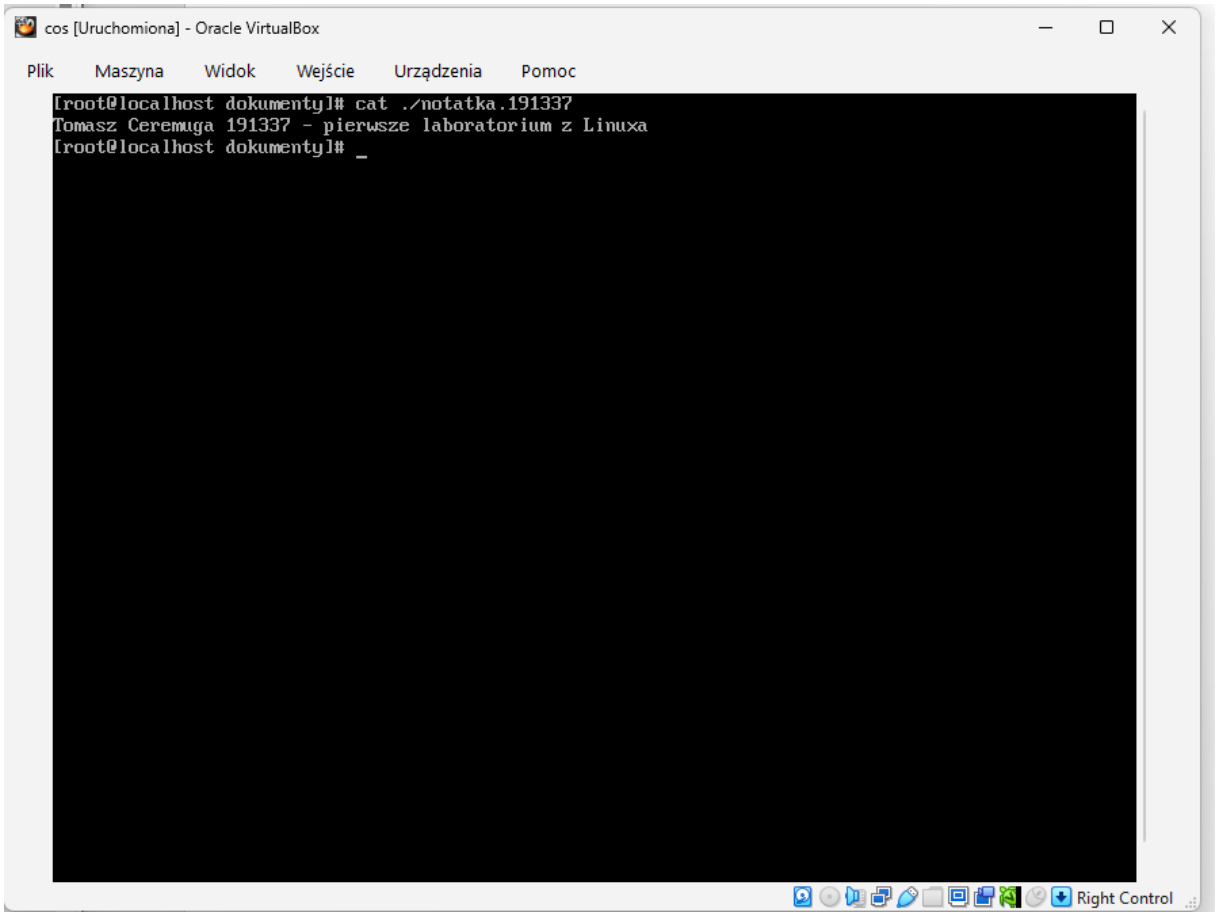
5.

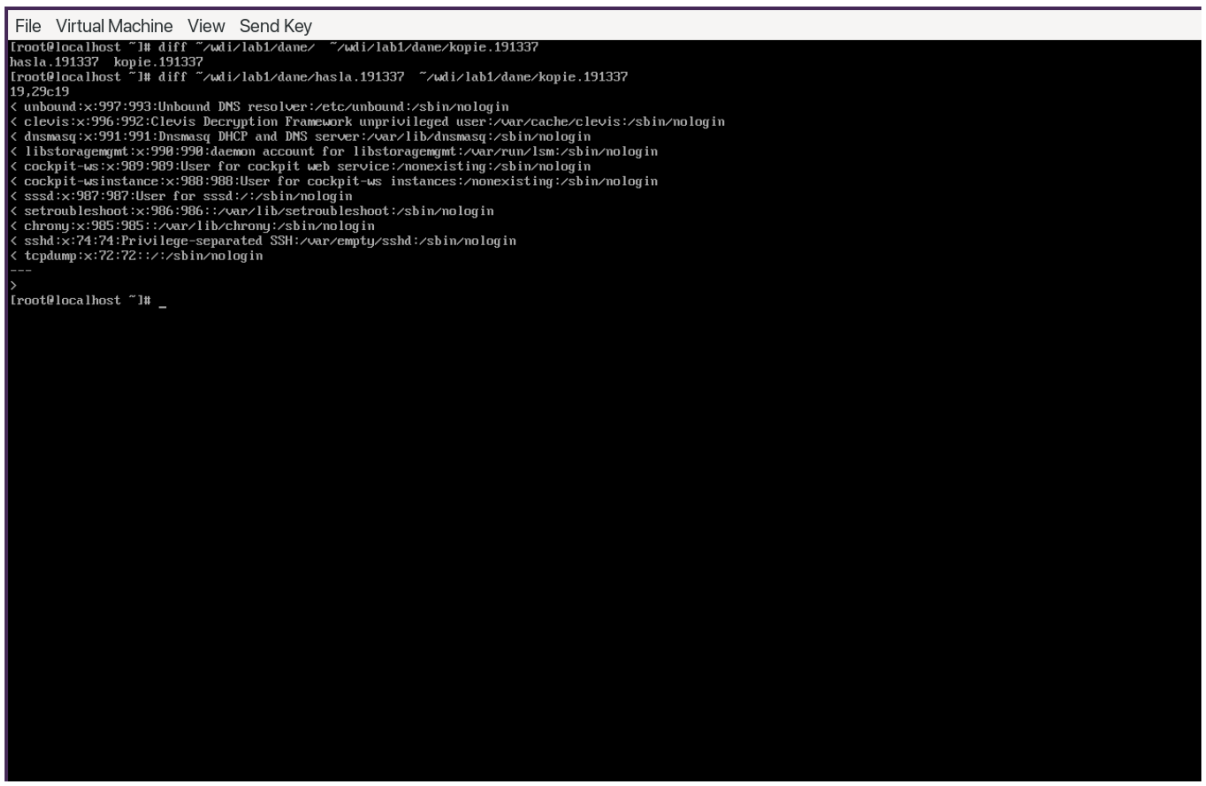

cos [Uruchomiona] - Oracle VirtualBox
Plik
Maszyna
Widok
Wejście
Urządzenia
Pomoc
[root@localhost ~]# mkdir -p wdi/lab1/{dane,dokumenty,programy}
[root@localhost ~]# ls wdi/lab1/
dane dokumenty programy
[root@localhost ~]# tree
.
├── anaconda-ks.cfg
└── wdi
    └── lab1
        ├── dane
        ├── dokumenty
        └── programy
5 directories, 1 file
[root@localhost ~]#
Right Control

6.

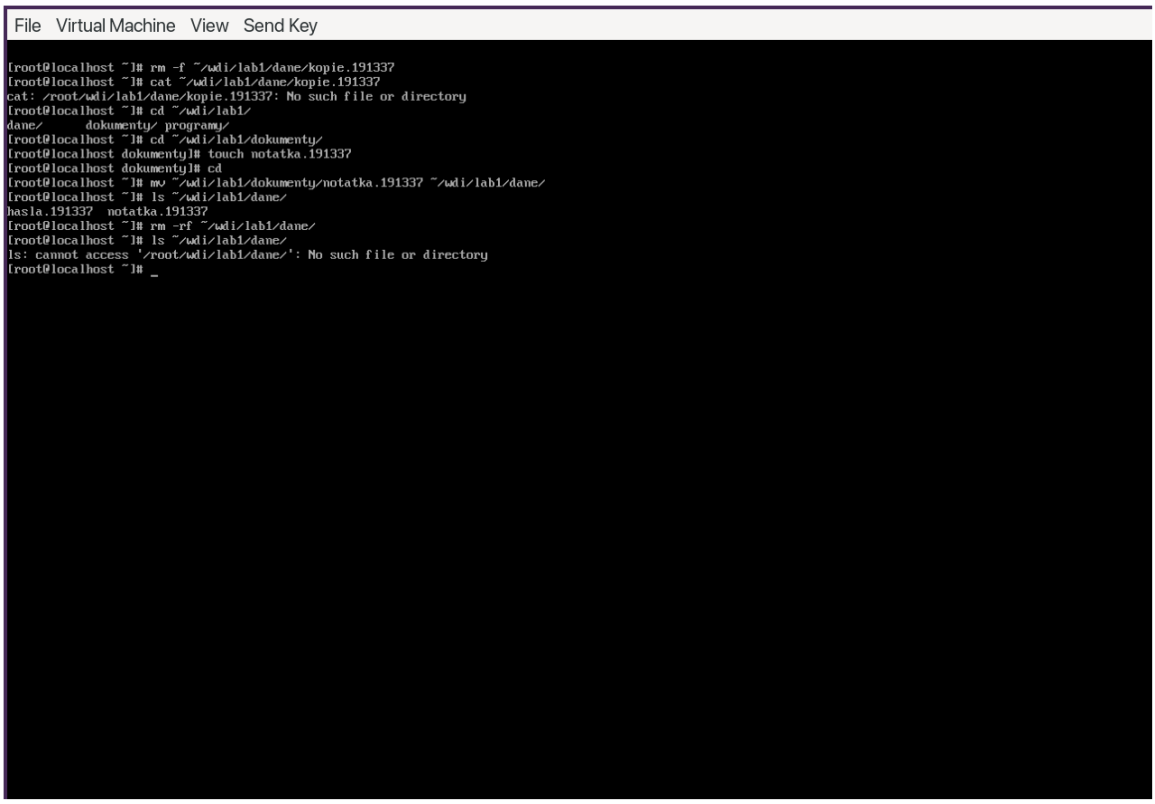


7. Tutaj wpisałem treść vimem i mi cały terminal wypełniło pustką dlatego zrobiłem tylko screena z cata



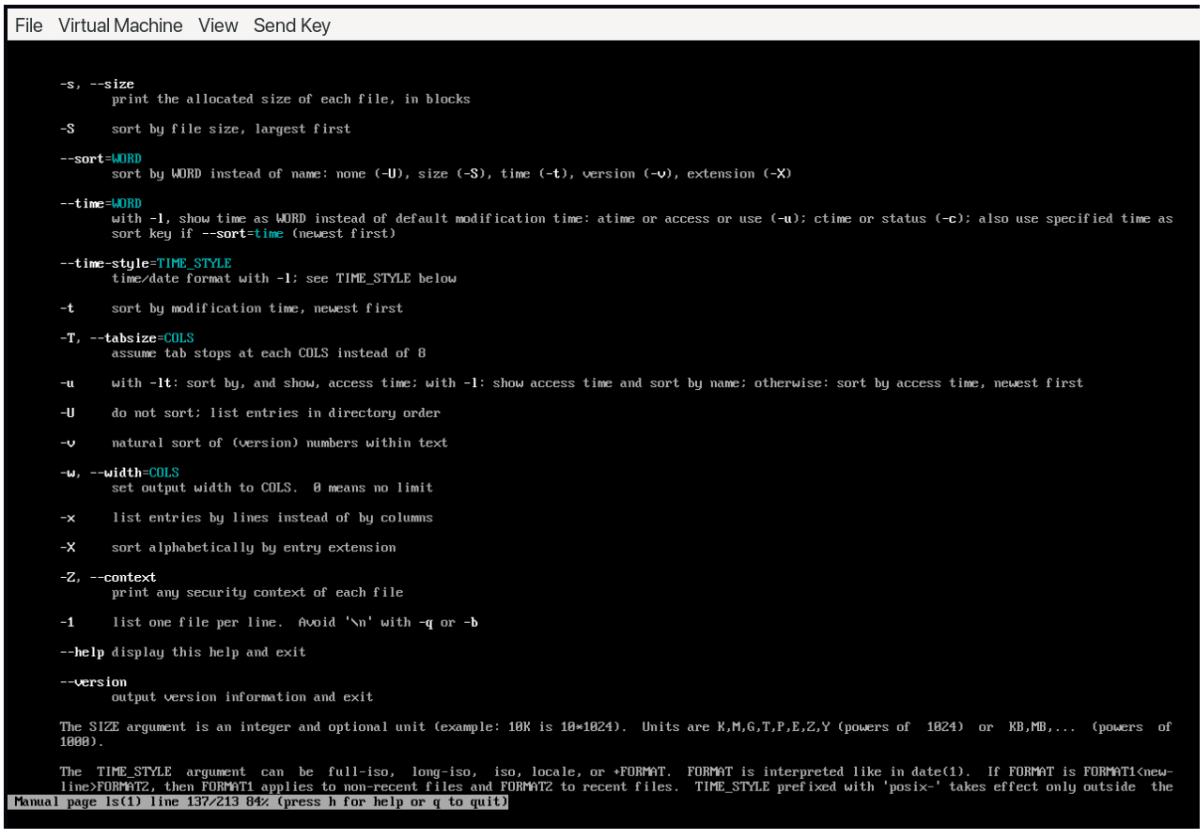
8. 

Jak widać usunięto kilka linijek

9. 

Tutaj przy próbie usunięcia niepustego katalogu otrzymamy ostrzeżenie że nie możemy tego zrobić bo znajdują się tam pliki
dlatego dodajemy flagę –r czyli usuń wraz z zawartością

Część 3

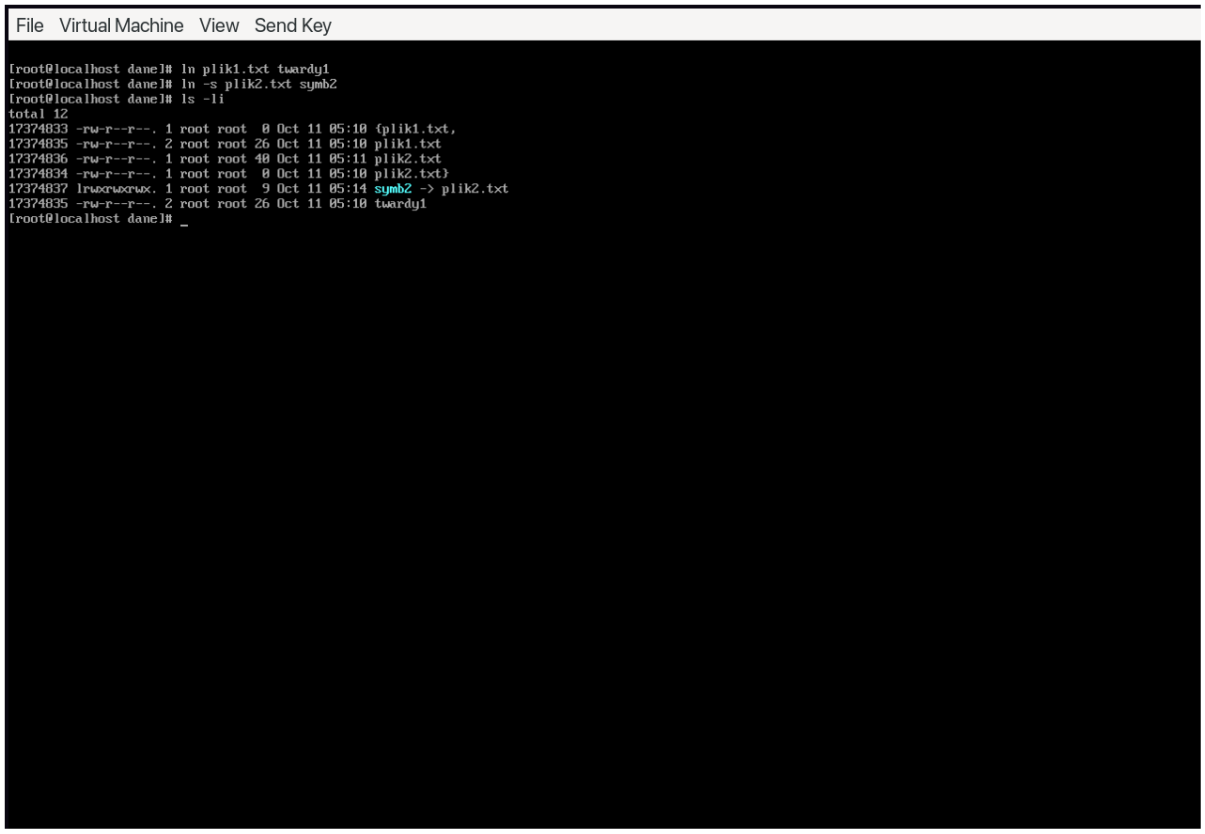


na górze, flaga –S da nam oczekiwane sortowanie

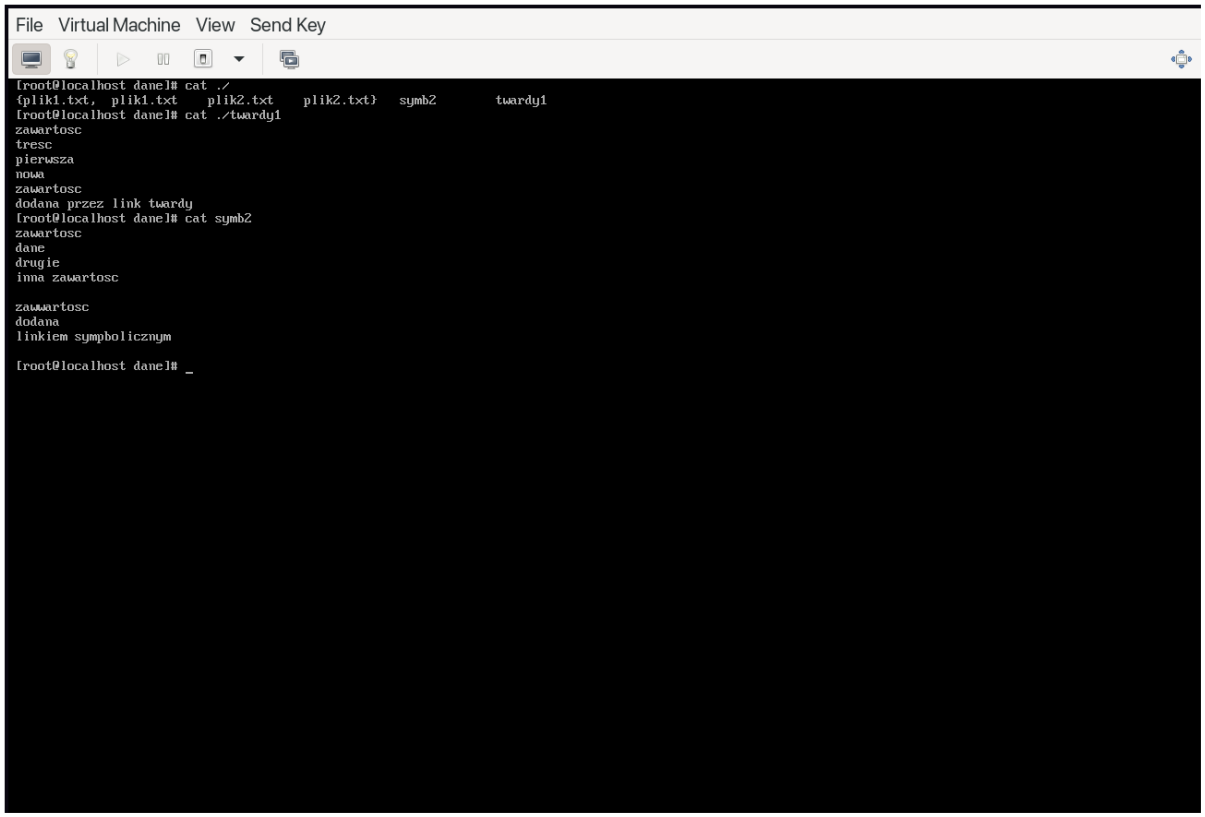
Również na górze
-R –r –recursive

Część 4

11.

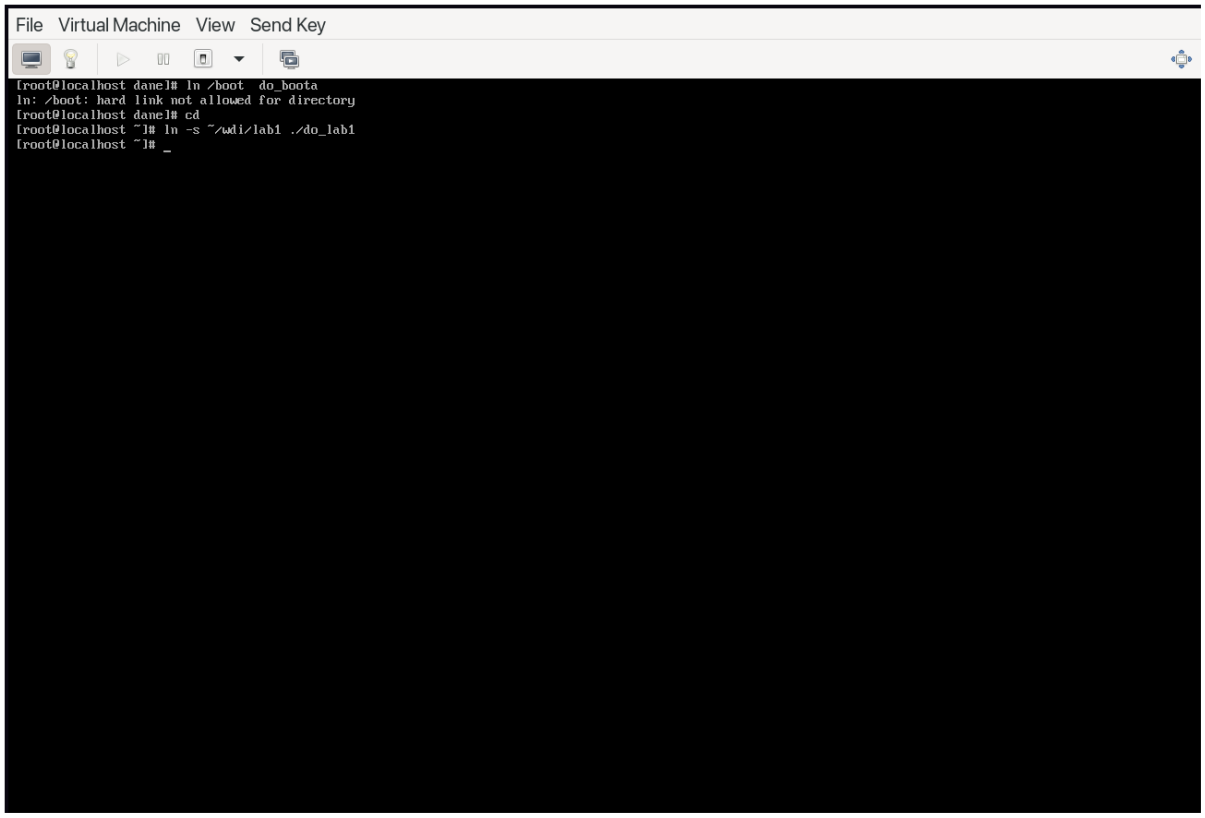


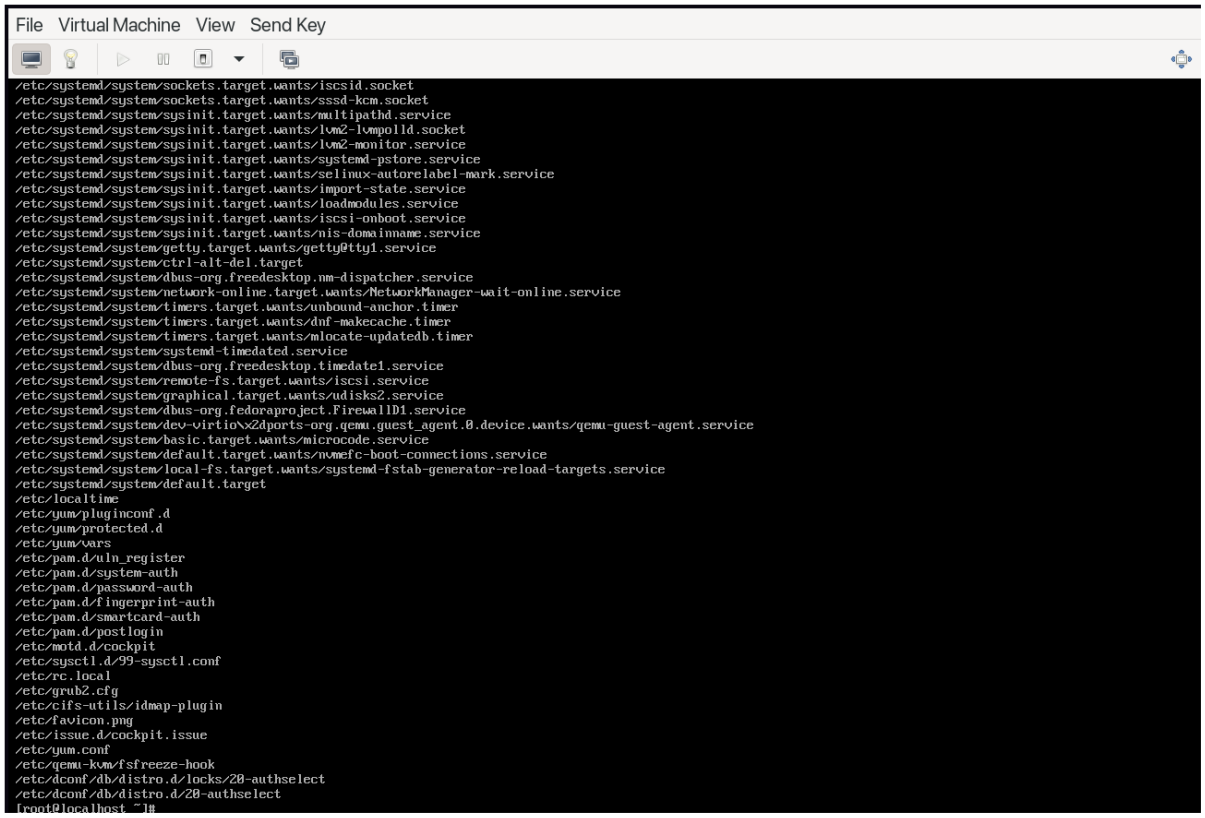
12.



Link twardy działa normalnie, bo wskazuje bezpośrednio na ten sam węzeł z danymi symboliczny nie działa, ma własny węzeł pamiętający tylko ścieżkę do usuniętego pliku

13.

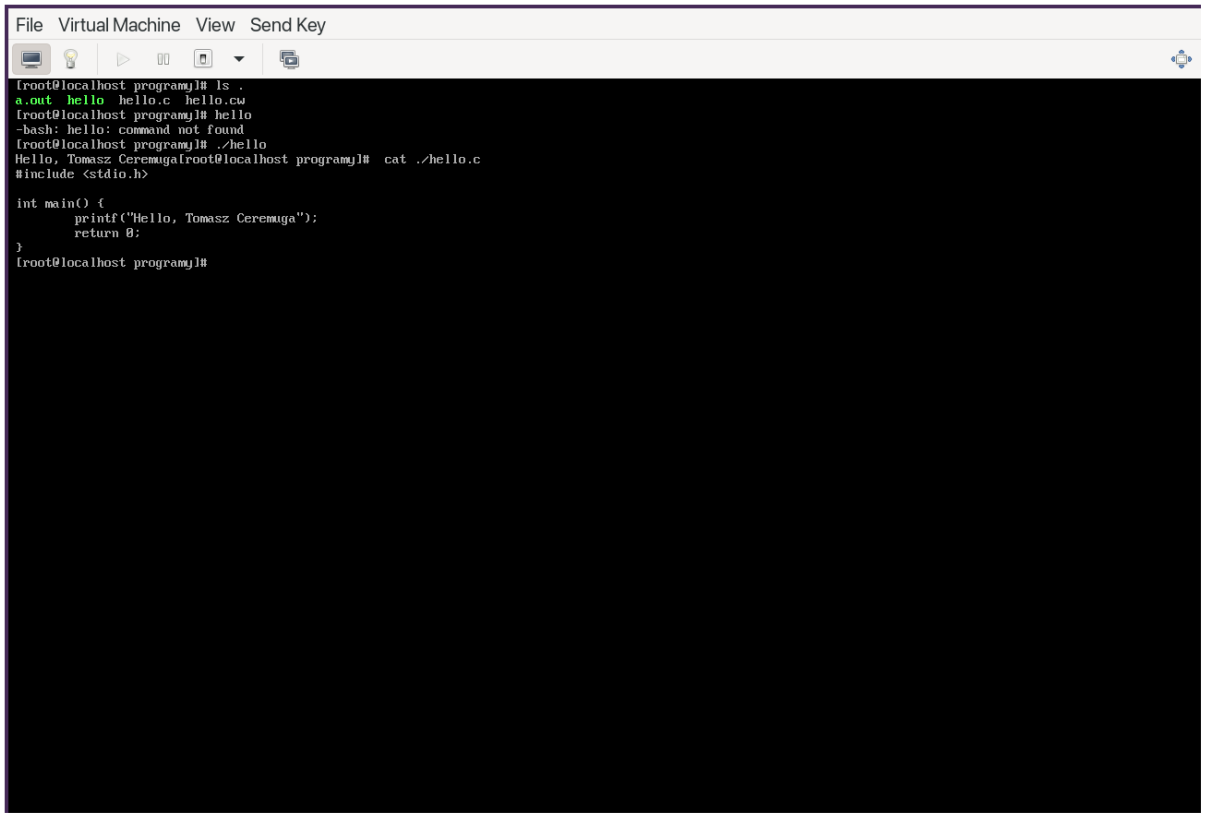




to są pliki skróty, wskazują na inny plik/ścieżkę

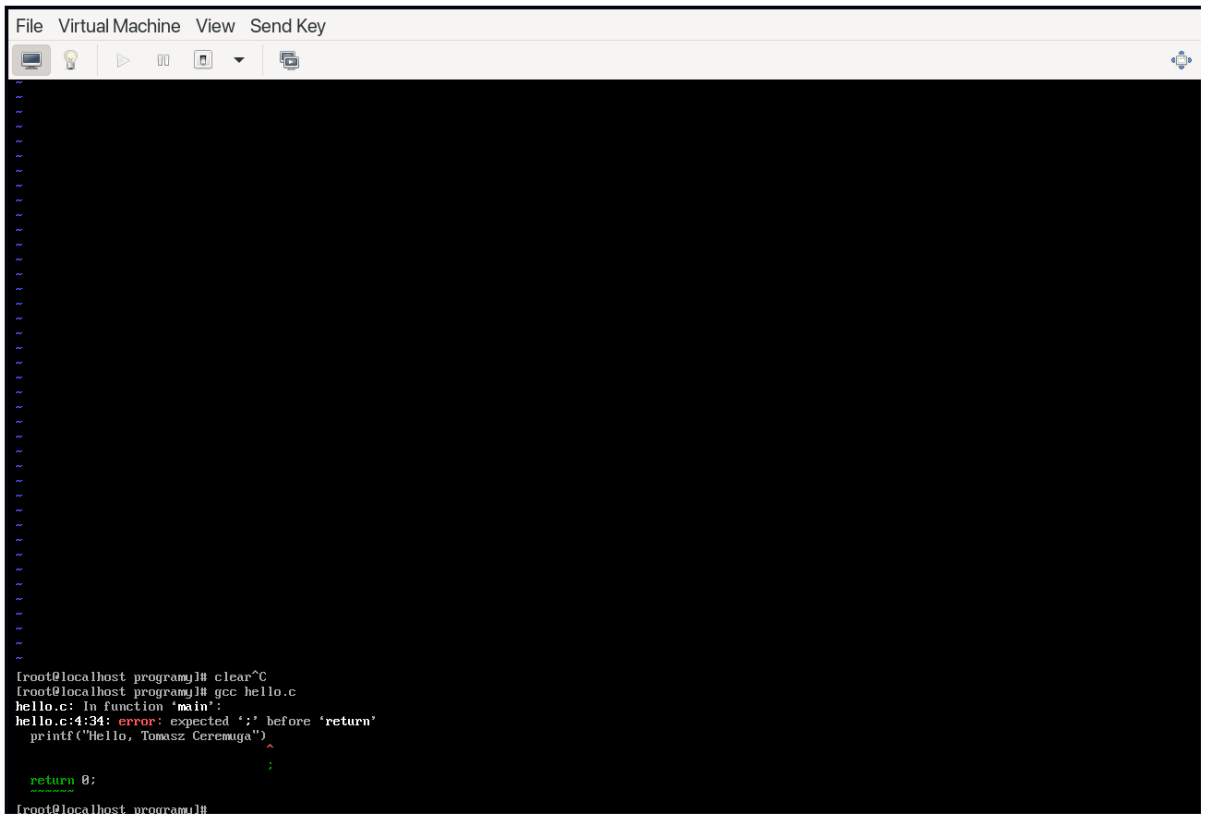
Część 5

14, 15



Hello odwołało by się do /bin natomiast my mamy plik wykonywalny u siebie

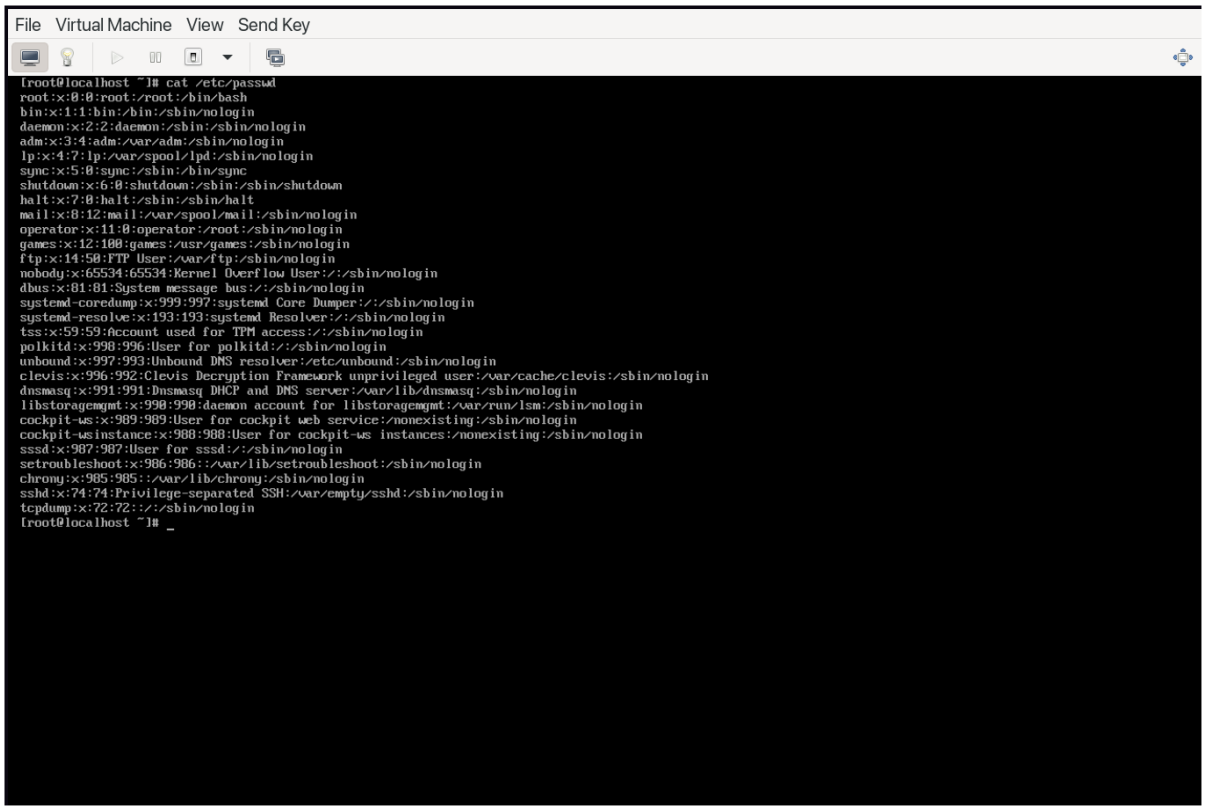
16.

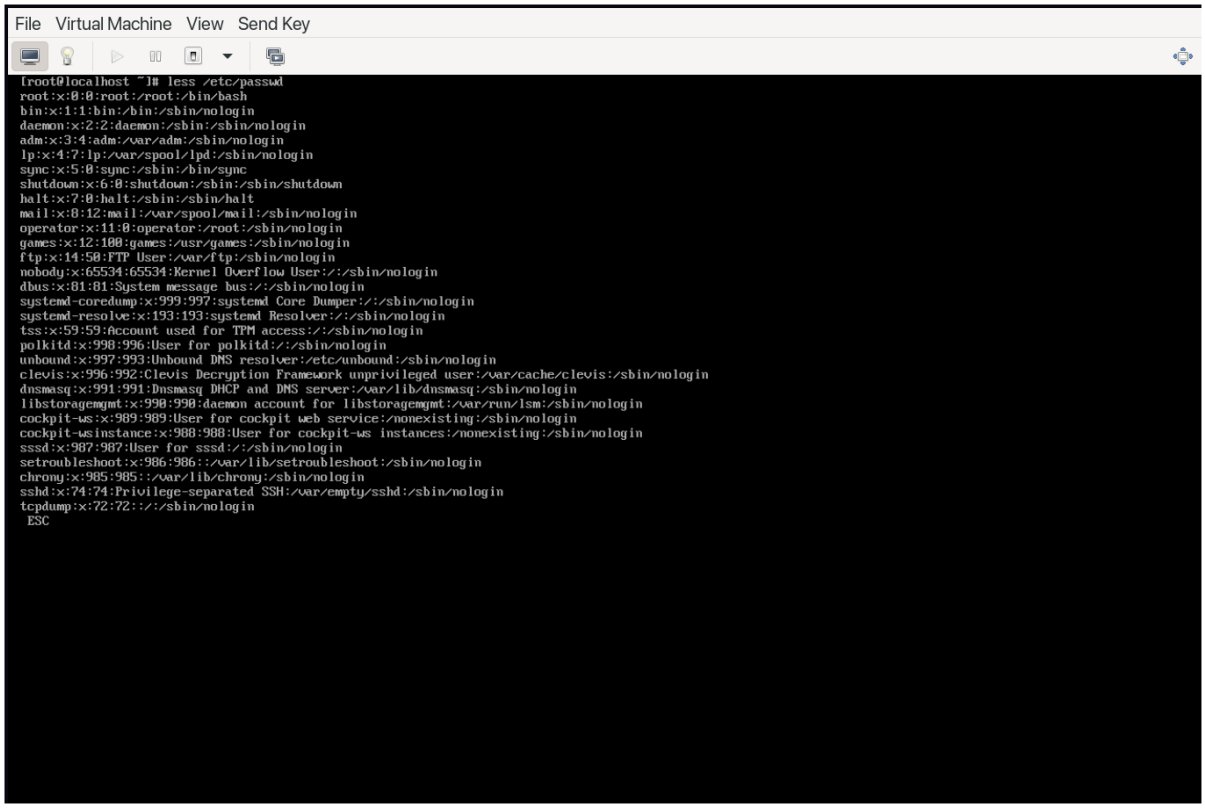


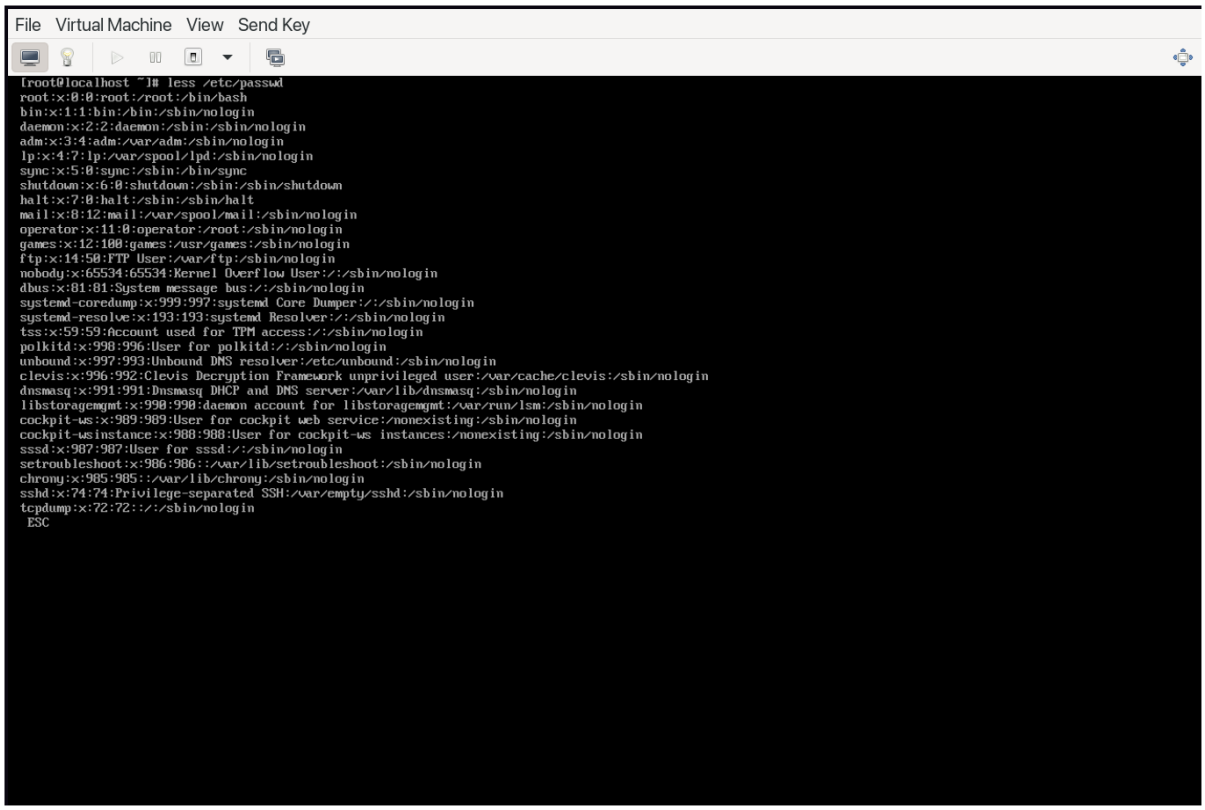
Oczekiwano ; przed rteturnem
plik hello.c linijka 4 znak 34

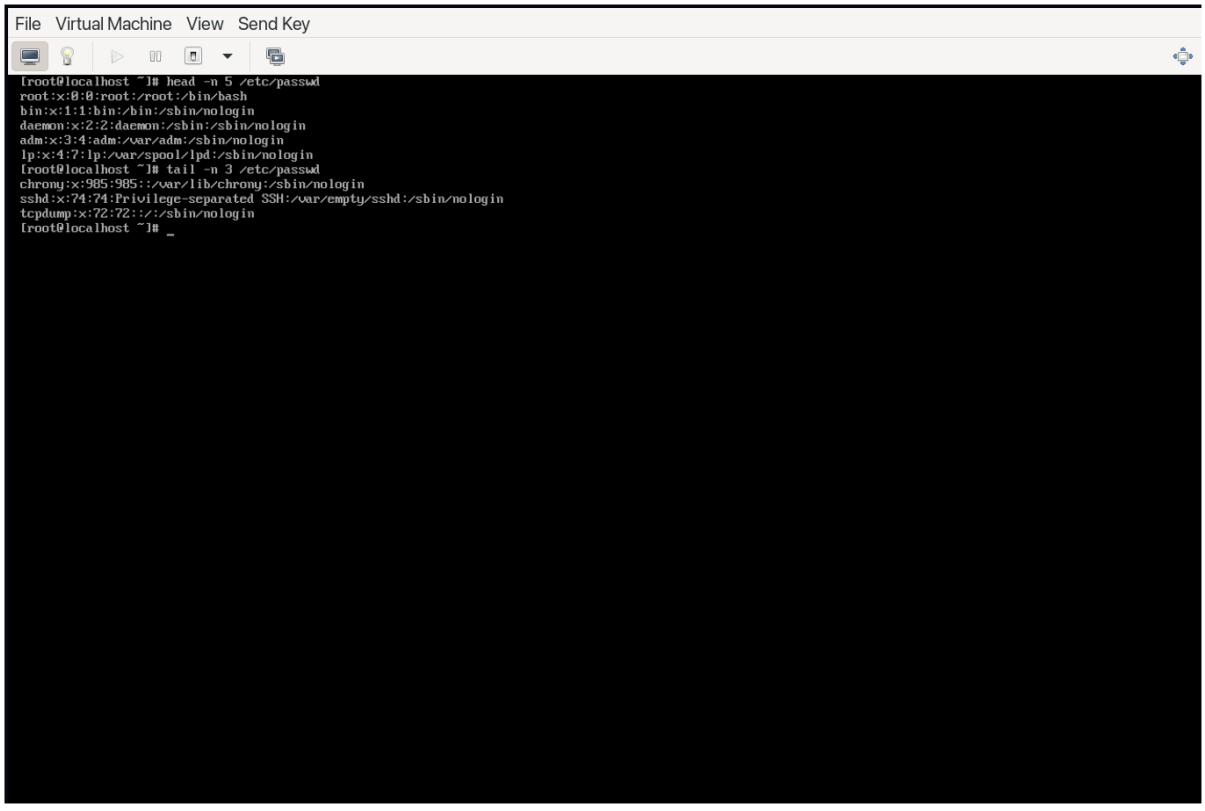
Lista 2

Część 1.


File Virtual Machine View Send Key
[root@localhost ~]# cat /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
systemd-coredump:x:999:997:systemd Core Dumper:/:/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/sbin/nologin
tss:x:59:59:Account used for TPM access:/:/sbin/nologin
polkitd:x:998:996:User for polkitd:/:/sbin/nologin
unbound:x:997:993:Unbound DNS resolver:/etc/unbound:/sbin/nologin
clevis:x:996:992:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/sbin/nologin
dnsmasq:x:991:991:Dnsmasq DHCP and DNS server:/var/lib/dnsmasq:/sbin/nologin
libstoragemgmt:x:990:990:daemon account for libstoragemgmt:/var/run/lsm:/sbin/nologin
cockpit-ws:x:989:989:User for cockpit web service:/nonexisting:/sbin/nologin
cockpit-wsinstance:x:988:988:User for cockpit-ws instances:/nonexisting:/sbin/nologin
sssd:x:987:987:User for sssd:/:/sbin/nologin
setroubleshoot:x:986:986::/var/lib/setroubleshoot:/sbin/nologin
chrony:x:985:985::/var/lib/chrony:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/var/empty/sshd:/sbin/nologin
tcpdump:x:72:72::/:/sbin/nologin
[root@localhost ~]# _

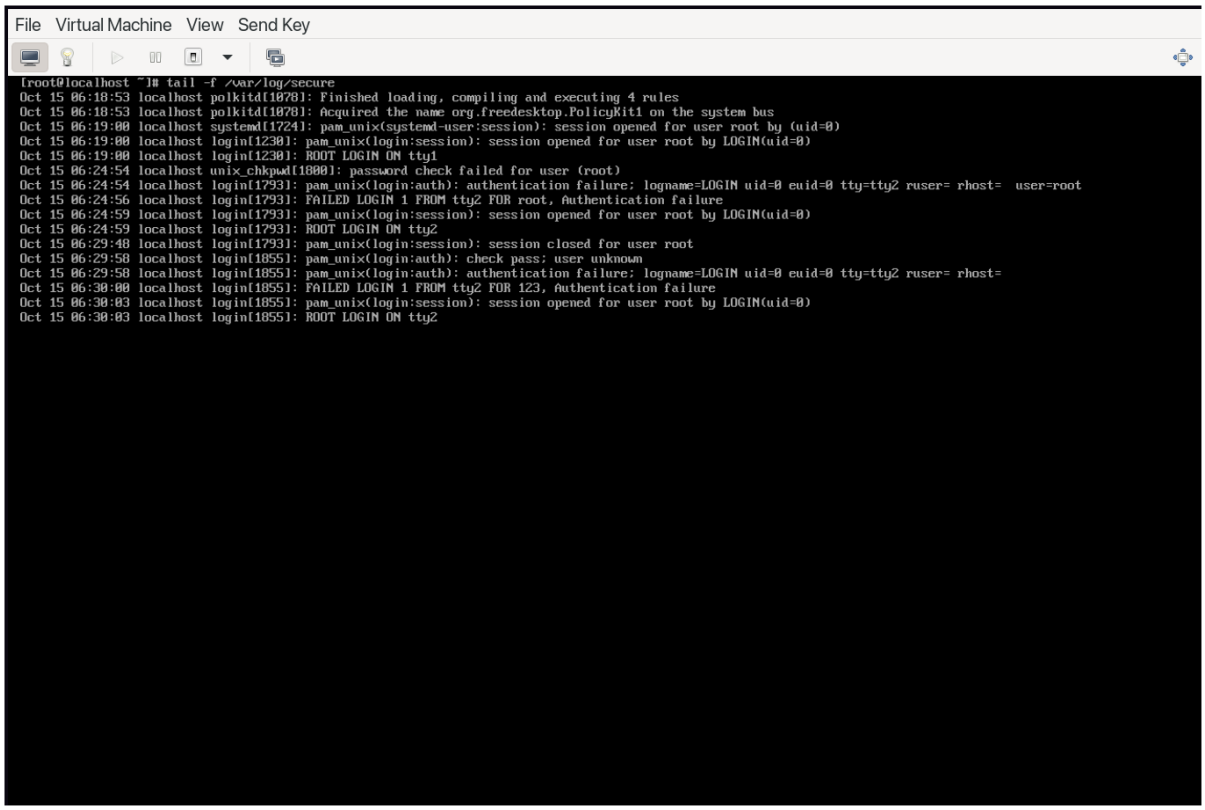

File Virtual Machine View Send Key
[root@localhost ~]# less /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
systemd-coredump:x:999:997:systemd Core Dumper:/:/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/sbin/nologin
tss:x:59:59:Account used for TPM access:/:/sbin/nologin
polkitd:x:998:996:User for polkitd:/:/sbin/nologin
unbound:x:997:993:Unbound DNS resolver:/etc/unbound:/sbin/nologin
clevis:x:996:992:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/sbin/nologin
dnsmasq:x:991:991:Dnsmasq DHCP and DNS server:/var/lib/dnsmasq:/sbin/nologin
libstoragemgmt:x:990:990:daemon account for libstoragemgmt:/var/run/lsm:/sbin/nologin
cockpit-ws:x:989:989:User for cockpit web service:/nonexisting:/sbin/nologin
cockpit-wsinstance:x:988:988:User for cockpit-ws instances:/nonexisting:/sbin/nologin
sssd:x:987:987:User for sssd:/:/sbin/nologin
setroubleshoot:x:986:986::/var/lib/setroubleshoot:/sbin/nologin
chrony:x:985:985::/var/lib/chrony:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/var/empty/sshd:/sbin/nologin
tcpdump:x:72:72::/:/sbin/nologin
ESC





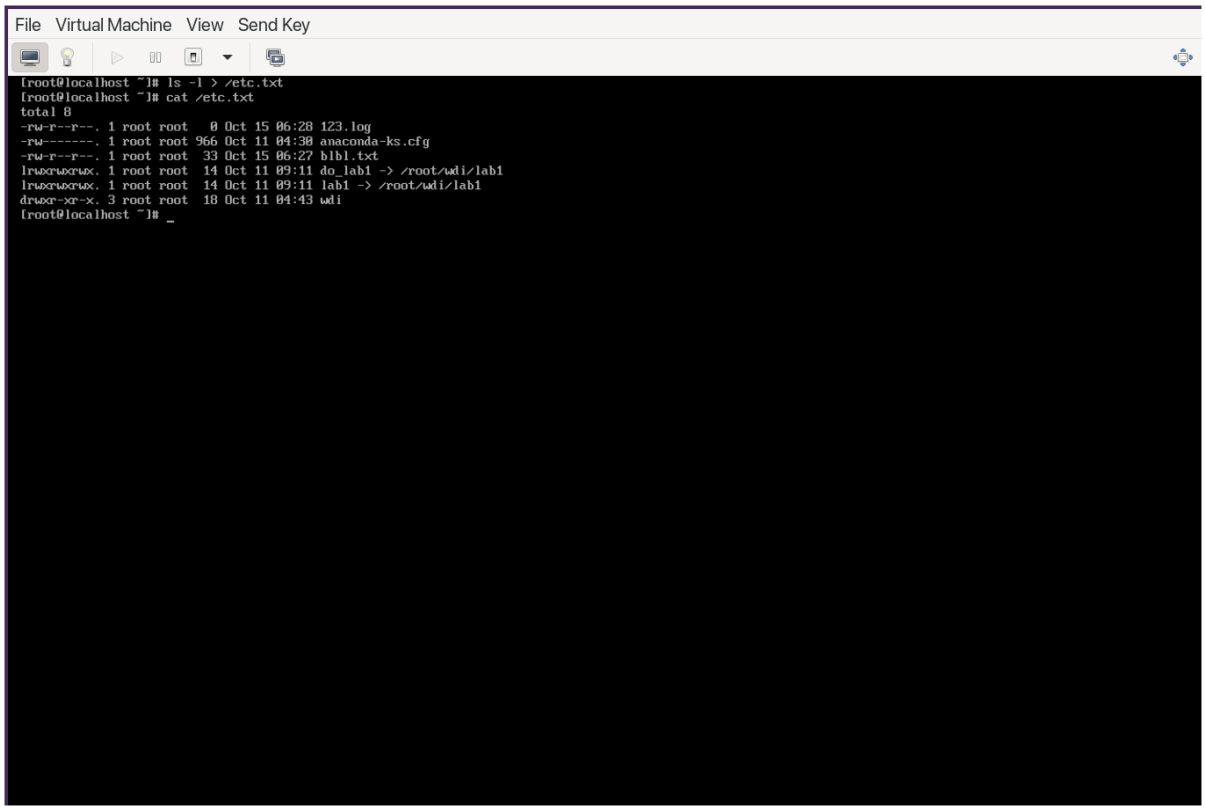
Less jest wygodniejsze kiedy plik jest długi i trzeba scrollować

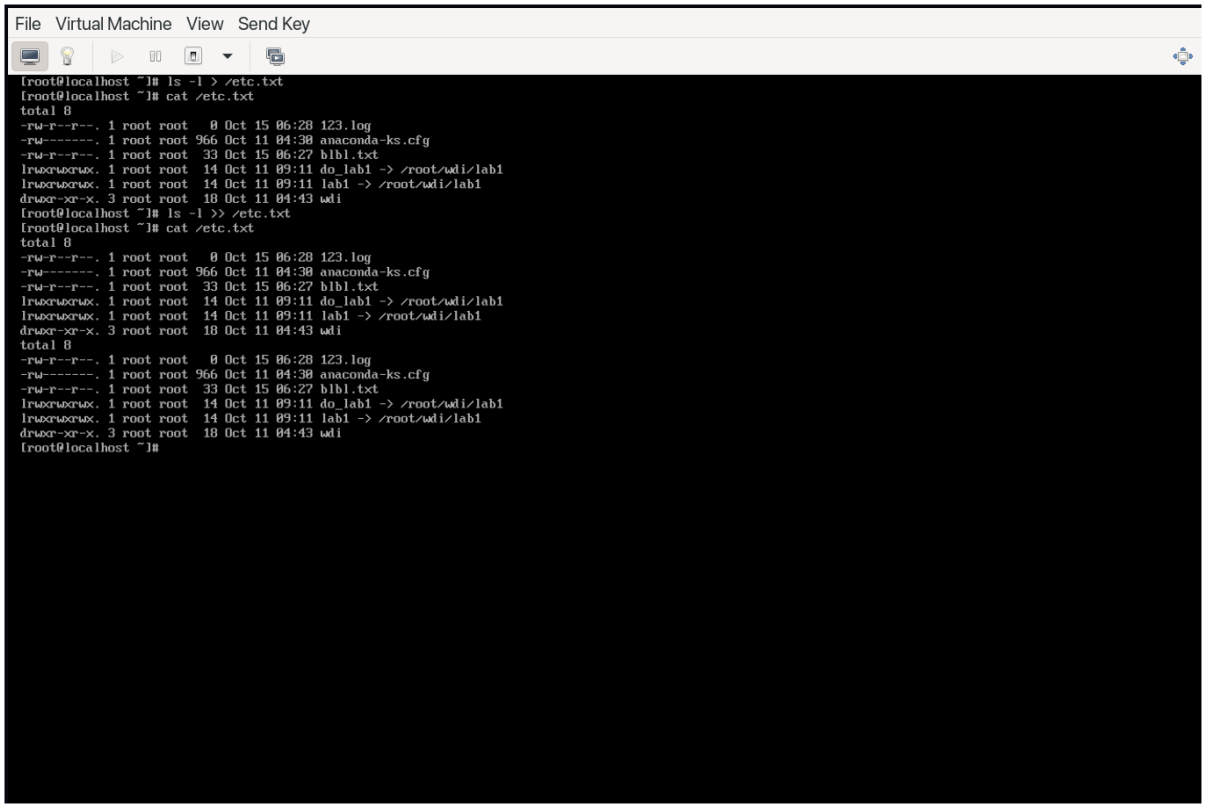
2.



-f na bierząco monitoruje ten plik
na screenie widać jak użytkownik się wylogował i zalogował

Część 2.

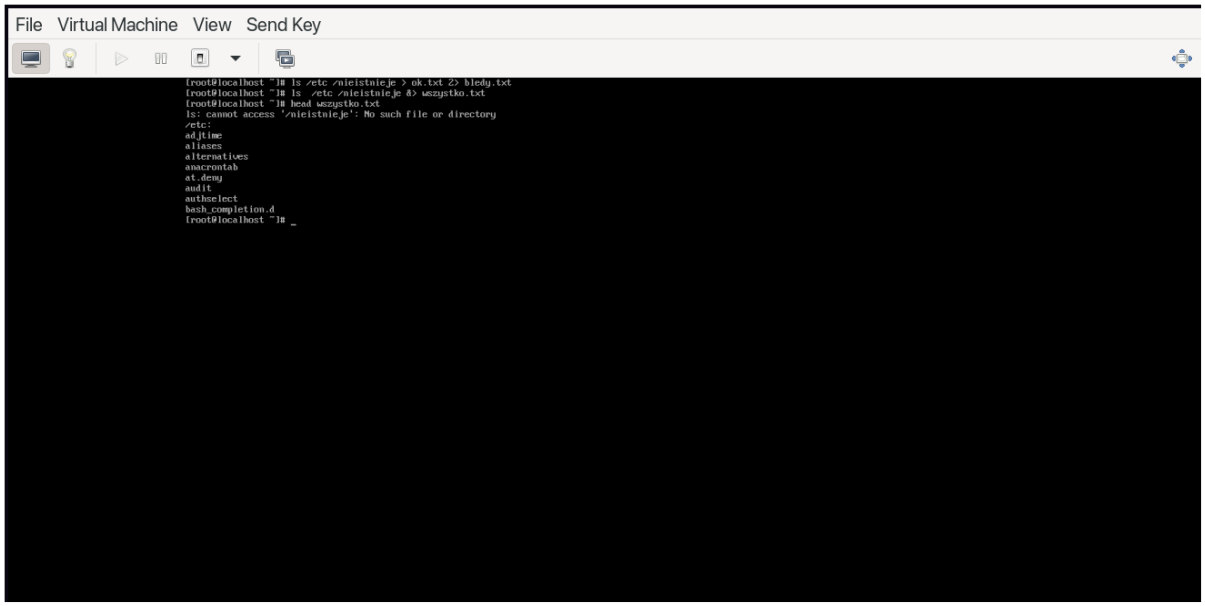




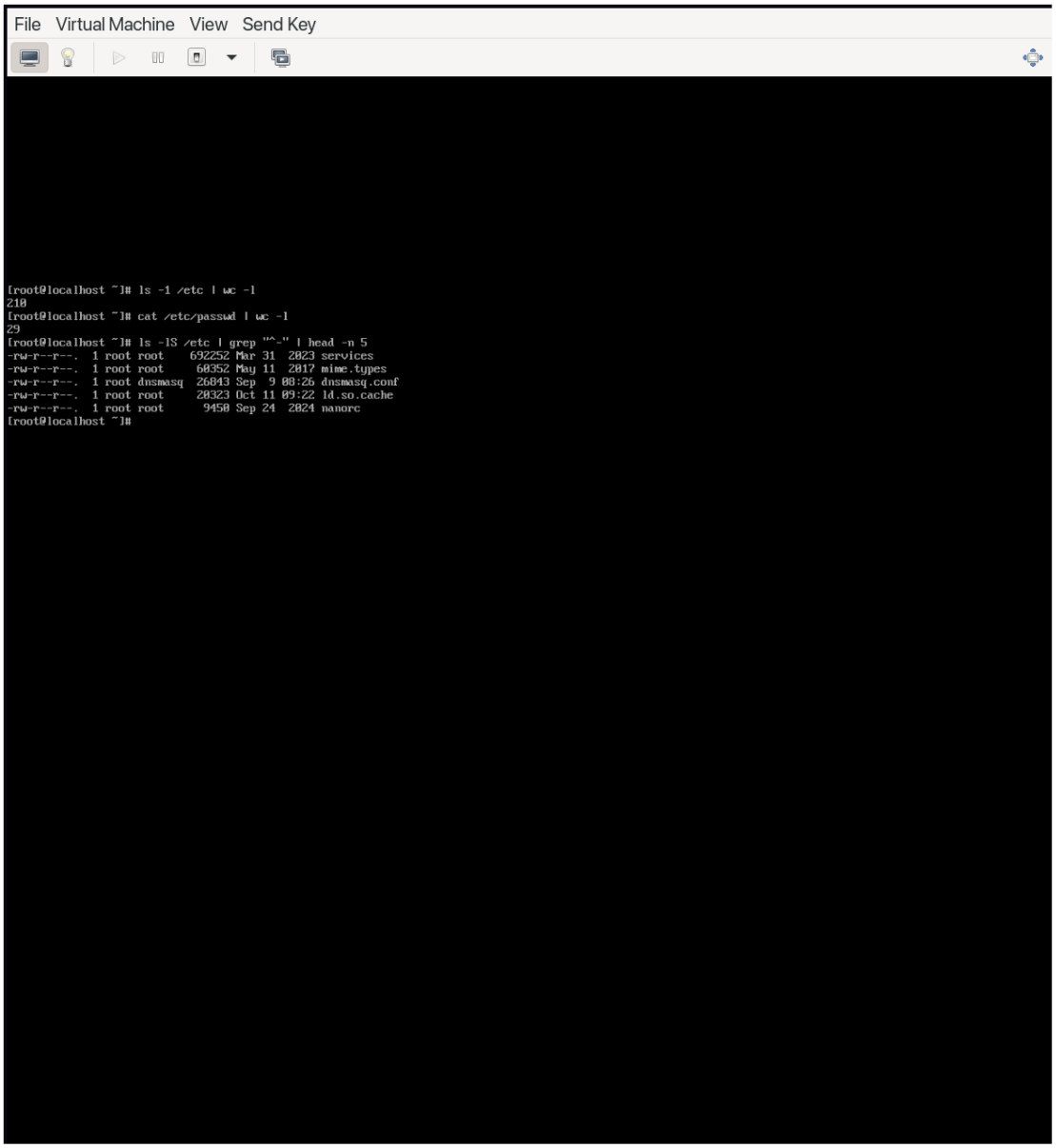
> nadpisuje

>> dopisuje

4.


File Virtual Machine View Send Key
[root@localhost ~]# ls /etc /nieistnieje > ok.txt 2> bledy.txt
[root@localhost ~]# ls /etc /nieistnieje &> wszystko.txt
[root@localhost ~]# head wszystko.txt
ls: cannot access '/nieistnieje': No such file or directory
/etc:
adjtime
aliases
alternatives
anacrontab
at.deny
audit
authselect
bash_completion.d
[root@localhost ~]#

5.


File Virtual Machine View Send Key
[root@localhost ~]# ls -1 /etc | wc -l
210
[root@localhost ~]# cat /etc/passwd | wc -l
29
[root@localhost ~]# ls -lS /etc | grep "^-" | head -n 5
-rw-r--r--. 1 root root 692252 Mar 31 2023 services
-rw-r--r--. 1 root root 60352 May 11 2017 mime.types
-rw-r--r--. 1 root dnsmasq 26043 Sep 9 00:26 dnsmasq.conf
-rw-r--r--. 1 root root 20323 Oct 11 09:22 ld.so.cache
-rw-r--r--. 1 root root 9450 Sep 24 2024 nanorc
[root@localhost ~]#

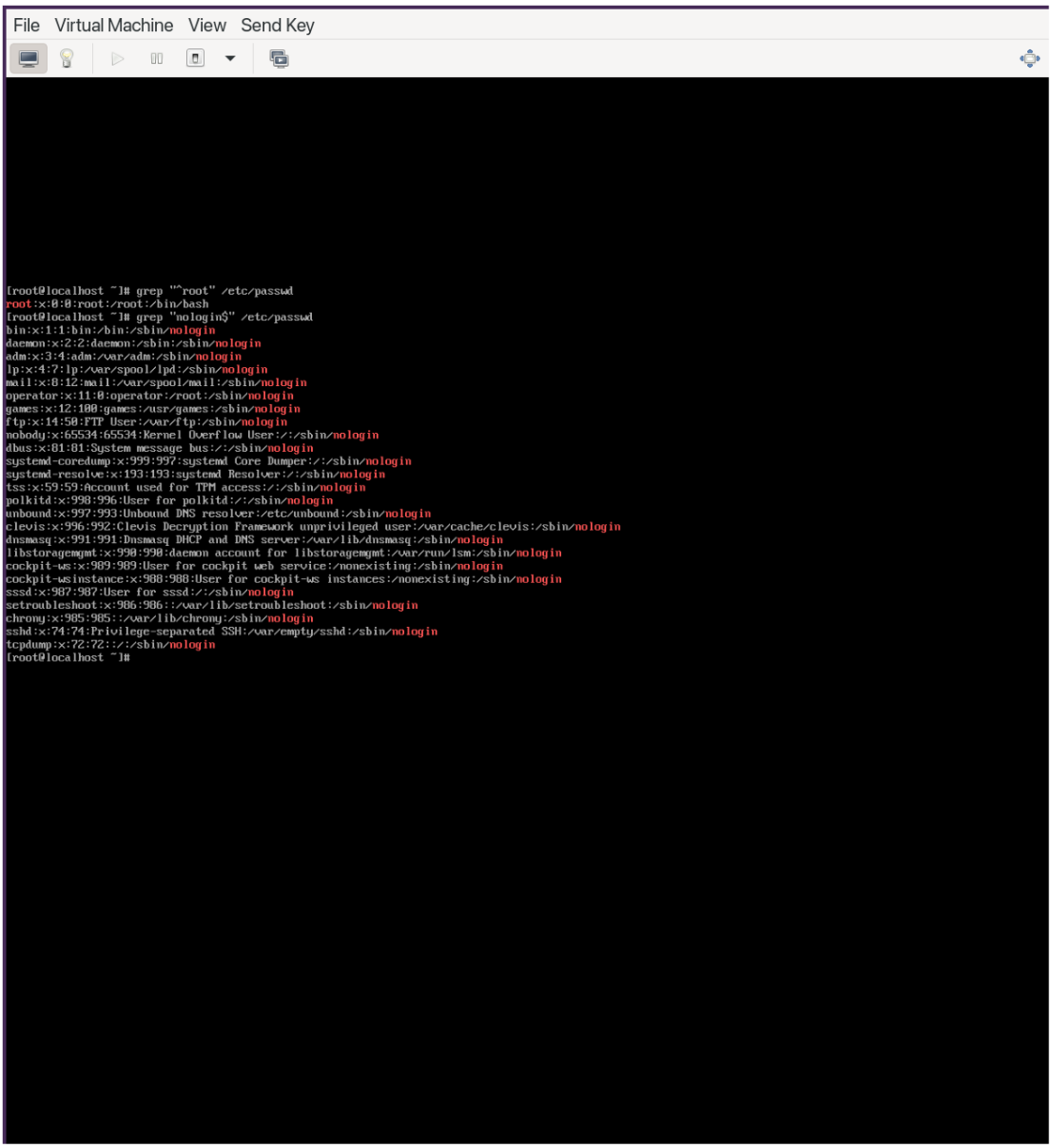


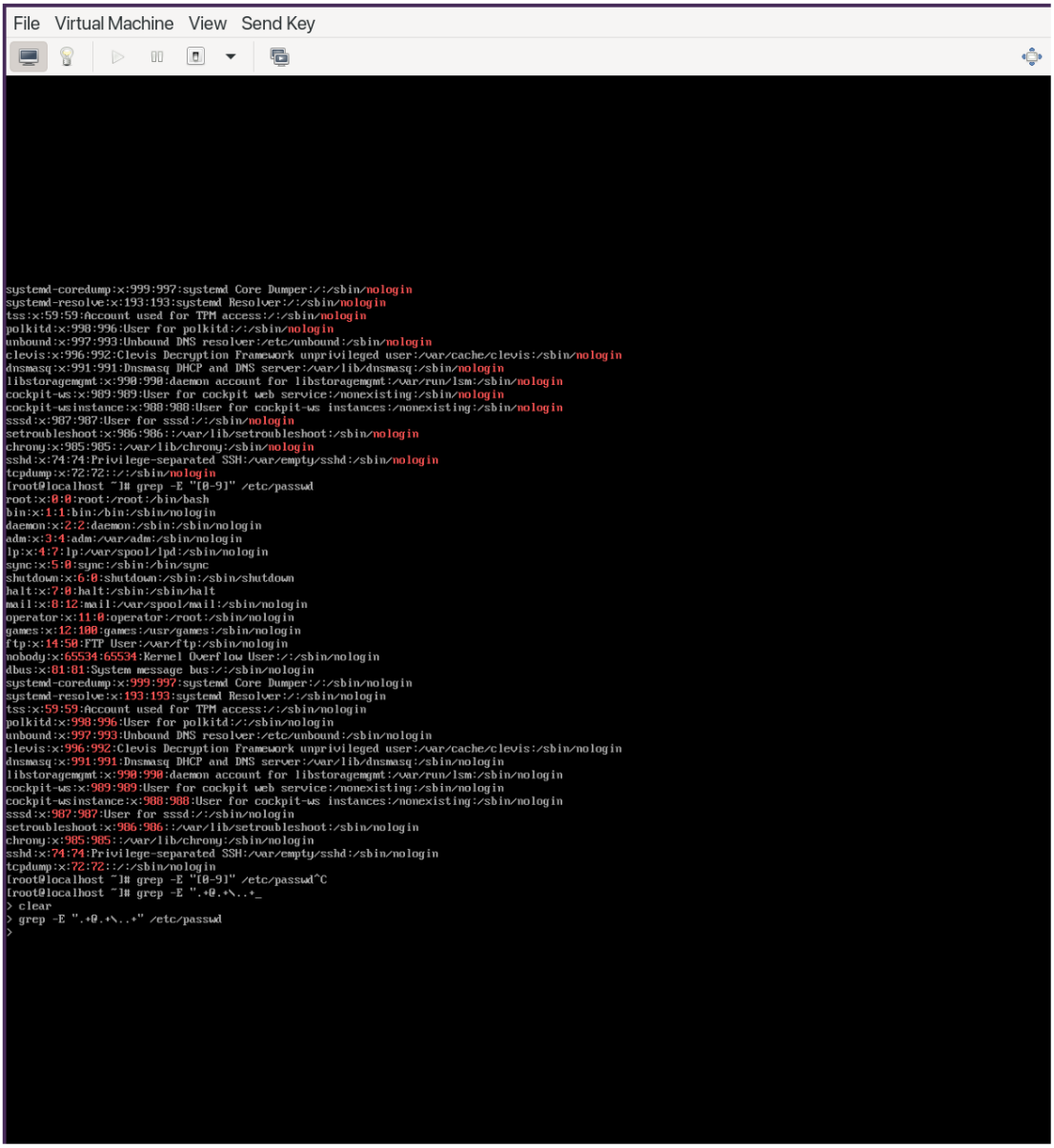
Jeden użytkownik ma /bin/bash

Część 3

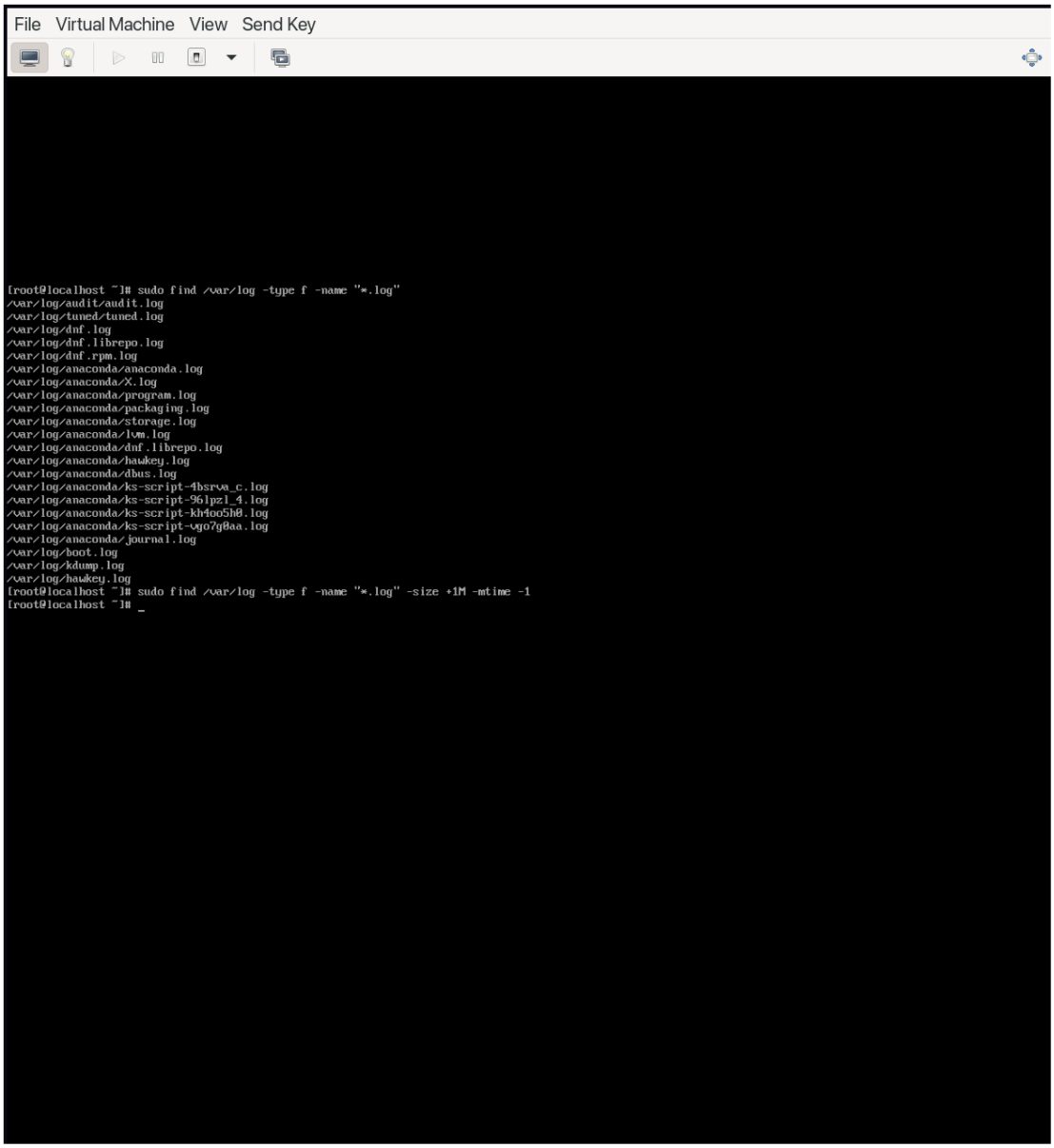
7. 
File Virtual Machine View Send Key
[root@localhost ~]# sudo grep -n "error" /var/log/messages
148:Oct 11 04:35:34 localhost kernel: unchecked MSR access error: WRMSR to 0x38f (tried to write 0x0001000f000000ff) at rIP: 0xffffffff8f89dbf4 (native_write_ms
r+0x4/0x20)
1534:Oct 11 08:46:13 localhost kernel: unchecked MSR access error: WRMSR to 0x38f (tried to write 0x0001000f000000ff) at rIP: 0xffffffffb3277294 (native_write_
sr+0x4/0x20)
2034:Oct 15 06:18:51 localhost kernel: unchecked MSR access error: WRMSR to 0x38f (tried to write 0x0001000f000000ff) at rIP: 0xffffffffbac77294 (native_write_
sr+0x4/0x20)
[root@localhost ~]# sudo grep -in "error" /var/log/messages
148:Oct 11 04:35:34 localhost kernel: unchecked MSR access error: WRMSR to 0x38f (tried to write 0x0001000f000000ff) at rIP: 0xffffffff8f89dbf4 (native_write_ms
r+0x4/0x20)
705:Oct 11 04:35:34 localhost kernel: RAS: Correctable Errors collector initialized.
1534:Oct 11 08:46:13 localhost kernel: unchecked MSR access error: WRMSR to 0x38f (tried to write 0x0001000f000000ff) at rIP: 0xffffffffb3277294 (native_write_
sr+0x4/0x20)
2034:Oct 15 06:18:51 localhost kernel: unchecked MSR access error: WRMSR to 0x38f (tried to write 0x0001000f000000ff) at rIP: 0xffffffffbac77294 (native_write_
sr+0x4/0x20)
[root@localhost ~]# sudo grep -c "error" /var/log/messages
3
[root@localhost ~]# sudo grep -ic "error" /var/log/messages
4
[root@localhost ~]#

8.


File Virtual Machine View Send Key
[root@localhost ~]# grep "^root" /etc/passwd
root:x:0:0:root:/root:/bin/bash
[root@localhost ~]# grep "nologin$" /etc/passwd
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
systemd-coredump:x:999:997:systemd Core Dumper:/:/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/sbin/nologin
tss:x:59:59:Account used for TPM access:/:/sbin/nologin
polkitd:x:998:996:User for polkitd:/:/sbin/nologin
unbound:x:997:993:Unbound DNS resolver:/etc/unbound:/sbin/nologin
clevis:x:996:992:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/sbin/nologin
dnsmasq:x:991:991:Dnsmasq DHCP and DNS server:/var/lib/dnsmasq:/sbin/nologin
libstoragemgmt:x:990:990:daemon account for libstoragemgmt:/var/run/lsm:/sbin/nologin
cockpit-ws:x:989:989:User for cockpit web service:/nonexisting:/sbin/nologin
cockpit-wsinstance:x:988:988:User for cockpit-ws instances:/nonexisting:/sbin/nologin
sssd:x:987:987:User for sssd:/:/sbin/nologin
setroubleshoot:x:986:986::/var/lib/setroubleshoot:/sbin/nologin
chrony:x:985:985::/var/lib/chrony:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/var/empty/sshd:/sbin/nologin
tcpdump:x:72:72::/:/sbin/nologin
[root@localhost ~]#


File Virtual Machine View Send Key
systemd-coredump:x:999:997:systemd Core Dumper:/:/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/sbin/nologin
tss:x:59:59:Account used for TPM access:/:/sbin/nologin
polkitd:x:998:996:User for polkitd:/:/sbin/nologin
unbound:x:997:993:Unbound DNS resolver:/etc/unbound:/sbin/nologin
clevis:x:996:992:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/sbin/nologin
dnsmasq:x:991:991:Dnsmasq DHCP and DNS server:/var/lib/dnsmasq:/sbin/nologin
libstoragemgmt:x:990:990:daemon account for libstoragemgmt:/var/run/lsm:/sbin/nologin
cockpit-ws:x:989:989:User for cockpit web service:/nonexisting:/sbin/nologin
cockpit-wsinstance:x:988:988:User for cockpit-ws instances:/nonexisting:/sbin/nologin
sssd:x:987:987:User for sssd:/:/sbin/nologin
setroubleshoot:x:986:986::/var/lib/setroubleshoot:/sbin/nologin
chrony:x:985:985::/var/lib/chrony:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/var/empty/sshd:/sbin/nologin
tcpdump:x:72:72::/:/sbin/nologin
[root@localhost ~]# grep -E "[0-9]" /etc/passwd
root:x:0:0:root:/root:/bin/bash
bin:x:1:1:bin:/bin:/sbin/nologin
daemon:x:2:2:daemon:/sbin:/sbin/nologin
adm:x:3:4:adm:/var/adm:/sbin/nologin
lp:x:4:7:lp:/var/spool/lpd:/sbin/nologin
sync:x:5:0:sync:/sbin:/bin/sync
shutdown:x:6:0:shutdown:/sbin:/sbin/shutdown
halt:x:7:0:halt:/sbin:/sbin/halt
mail:x:8:12:mail:/var/spool/mail:/sbin/nologin
operator:x:11:0:operator:/root:/sbin/nologin
games:x:12:100:games:/usr/games:/sbin/nologin
ftp:x:14:50:FTP User:/var/ftp:/sbin/nologin
nobody:x:65534:65534:Kernel Overflow User:/:/sbin/nologin
dbus:x:81:81:System message bus:/:/sbin/nologin
systemd-coredump:x:999:997:systemd Core Dumper:/:/sbin/nologin
systemd-resolve:x:193:193:systemd Resolver:/:/sbin/nologin
tss:x:59:59:Account used for TPM access:/:/sbin/nologin
polkitd:x:998:996:User for polkitd:/:/sbin/nologin
unbound:x:997:993:Unbound DNS resolver:/etc/unbound:/sbin/nologin
clevis:x:996:992:Clevis Decryption Framework unprivileged user:/var/cache/clevis:/sbin/nologin
dnsmasq:x:991:991:Dnsmasq DHCP and DNS server:/var/lib/dnsmasq:/sbin/nologin
libstoragemgmt:x:990:990:daemon account for libstoragemgmt:/var/run/lsm:/sbin/nologin
cockpit-ws:x:989:989:User for cockpit web service:/nonexisting:/sbin/nologin
cockpit-wsinstance:x:988:988:User for cockpit-ws instances:/nonexisting:/sbin/nologin
sssd:x:987:987:User for sssd:/:/sbin/nologin
setroubleshoot:x:986:986::/var/lib/setroubleshoot:/sbin/nologin
chrony:x:985:985::/var/lib/chrony:/sbin/nologin
sshd:x:74:74:Privilege-separated SSH:/var/empty/sshd:/sbin/nologin
tcpdump:x:72:72::/:/sbin/nologin
[root@localhost ~]# grep -E "[0-9]" /etc/passwd^C
[root@localhost ~]# grep -E ".+@.+\..+_
> clear
> grep -E ".+@.+\..+" /etc/passwd
>


File Virtual Machine View Send Key
[root@localhost ~]# grep -rl "printf" ~/wdi
/root/wdi/lab1/programy/a.out
/root/wdi/lab1/programy/hello
/root/wdi/lab1/programy/hello.c
(arg: 2)

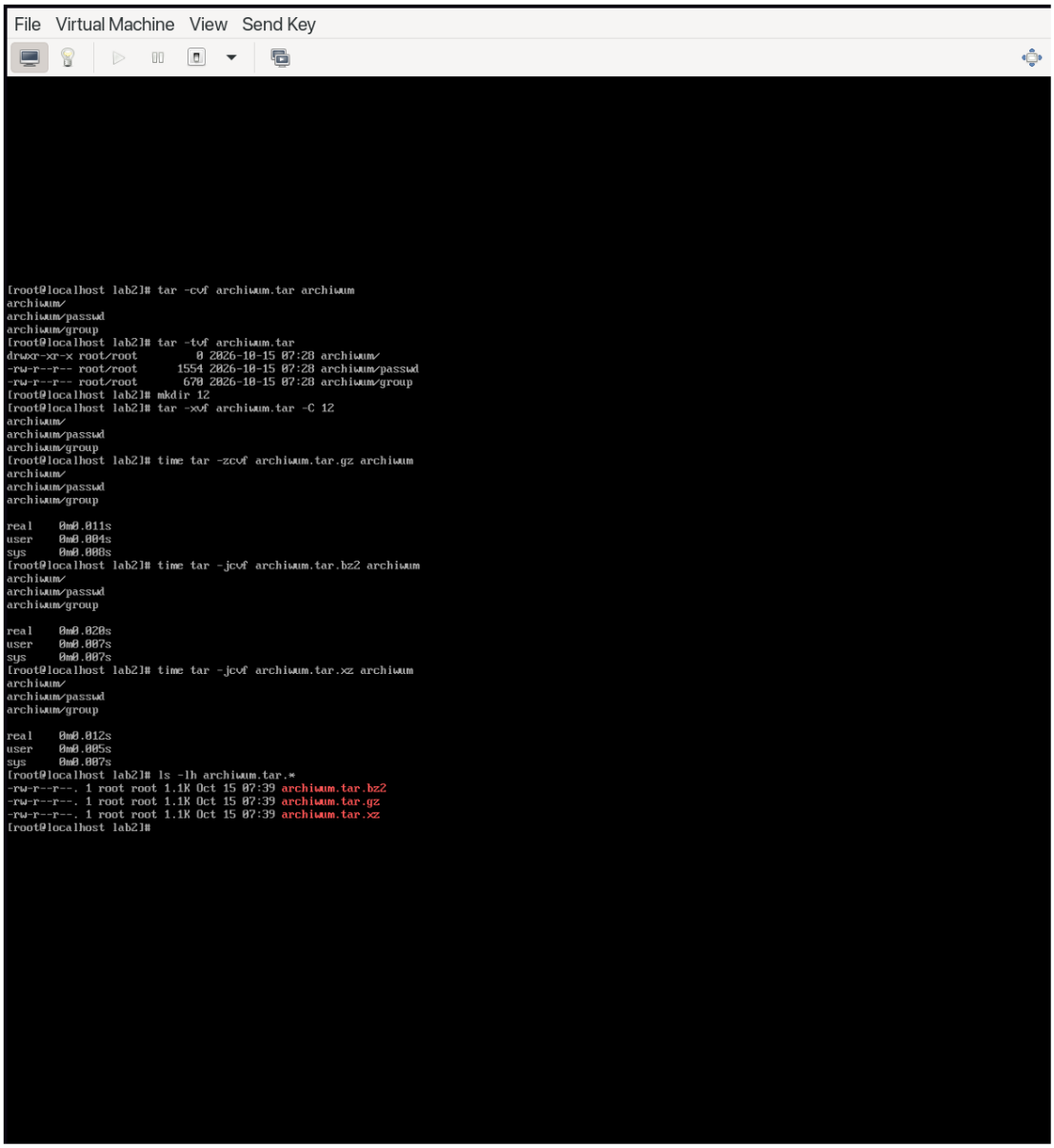

File Virtual Machine View Send Key
[root@localhost ~]# sudo find /var/log -type f -name "*.log"
/var/log/audit/audit.log
/var/log/tuned/tuned.log
/var/log/dnf.log
/var/log/dnf.librepo.log
/var/log/dnf.rpm.log
/var/log/anaconda/anaconda.log
/var/log/anaconda/X.log
/var/log/anaconda/program.log
/var/log/anaconda/packaging.log
/var/log/anaconda/storage.log
/var/log/anaconda/lvm.log
/var/log/anaconda/dnf.librepo.log
/var/log/anaconda/hawkey.log
/var/log/anaconda/dbus.log
/var/log/anaconda/ks-script-4bsrva_c.log
/var/log/anaconda/ks-script-96lpzl_4.log
/var/log/anaconda/ks-script-kh4oo5h0.log
/var/log/anaconda/ks-script-vgo7g0aa.log
/var/log/anaconda/journal.log
/var/log/boot.log
/var/log/kdump.log
/var/log/hawkey.log
[root@localhost ~]# sudo find /var/log -type f -name "*.log" -size +1M -mtime -1
[root@localhost ~]# _

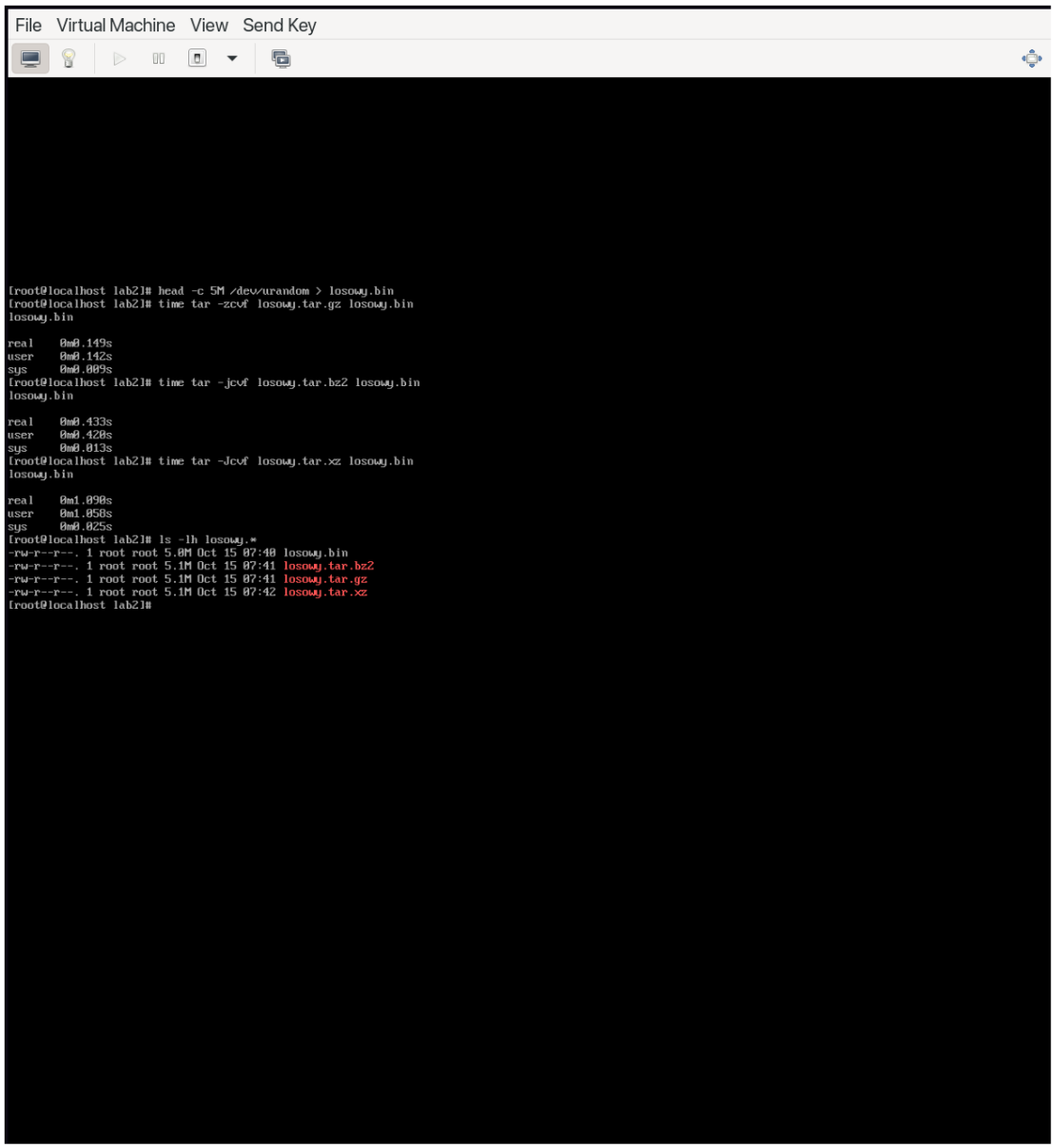
11.


File Virtual Machine View Send Key
[root@localhost ~]# find ~ -type f -name "*.c" -exec wc -l {} \;
6 /root/wdi/lab1/programy/hello.c
[root@localhost ~]#


File Virtual Machine View Send Key
[root@localhost lab2]# tar -cvf archiwum.tar archiwum
archiwum/
archiwum/passwd
archiwum/group
[root@localhost lab2]# tar -tvf archiwum.tar
drwxr-xr-x root/root 0 2026-10-15 07:28 archiwum/
-rw-r--r-- root/root 1554 2026-10-15 07:28 archiwum/passwd
-rw-r--r-- root/root 670 2026-10-15 07:28 archiwum/group
[root@localhost lab2]# mkdir 12
[root@localhost lab2]# tar -xvf archiwum.tar -C 12
archiwum/
archiwum/passwd
archiwum/group
[root@localhost lab2]#

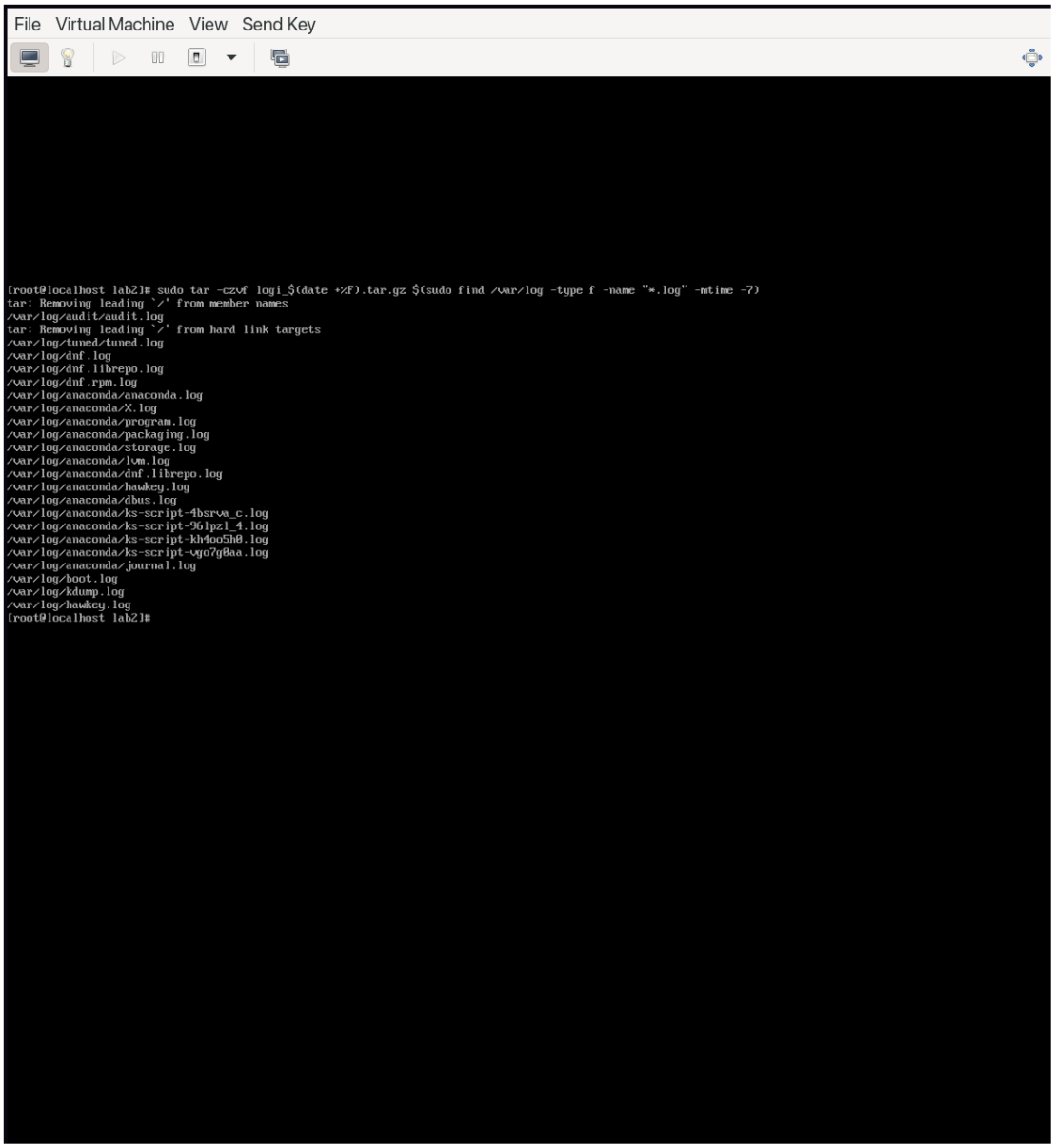
13


File Virtual Machine View Send Key
[root@localhost lab2]# tar -cvf archiwum.tar archiwum
archiwum/
archiwum/passwd
archiwum/group
[root@localhost lab2]# tar -tvf archiwum.tar
drwxr-xr-x root/root 0 2026-10-15 07:28 archiwum/
-rw-r--r-- root/root 1554 2026-10-15 07:28 archiwum/passwd
-rw-r--r-- root/root 670 2026-10-15 07:28 archiwum/group
[root@localhost lab2]# mkdir 12
[root@localhost lab2]# tar -xvf archiwum.tar -C 12
archiwum/
archiwum/passwd
archiwum/group
[root@localhost lab2]# time tar -zcvf archiwum.tar.gz archiwum
archiwum/
archiwum/passwd
archiwum/group
real 0m0.011s
user 0m0.004s
sys 0m0.008s
[root@localhost lab2]# time tar -jcvf archiwum.tar.bz2 archiwum
archiwum/
archiwum/passwd
archiwum/group
real 0m0.020s
user 0m0.007s
sys 0m0.007s
[root@localhost lab2]# time tar -jcvf archiwum.tar.xz archiwum
archiwum/
archiwum/passwd
archiwum/group
real 0m0.012s
user 0m0.005s
sys 0m0.007s
[root@localhost lab2]# ls -lh archiwum.tar.*
-rw-r--r--. 1 root root 1.1K Oct 15 07:39 archiwum.tar.bz2
-rw-r--r--. 1 root root 1.1K Oct 15 07:39 archiwum.tar.gz
-rw-r--r--. 1 root root 1.1K Oct 15 07:39 archiwum.tar.xz
[root@localhost lab2]#



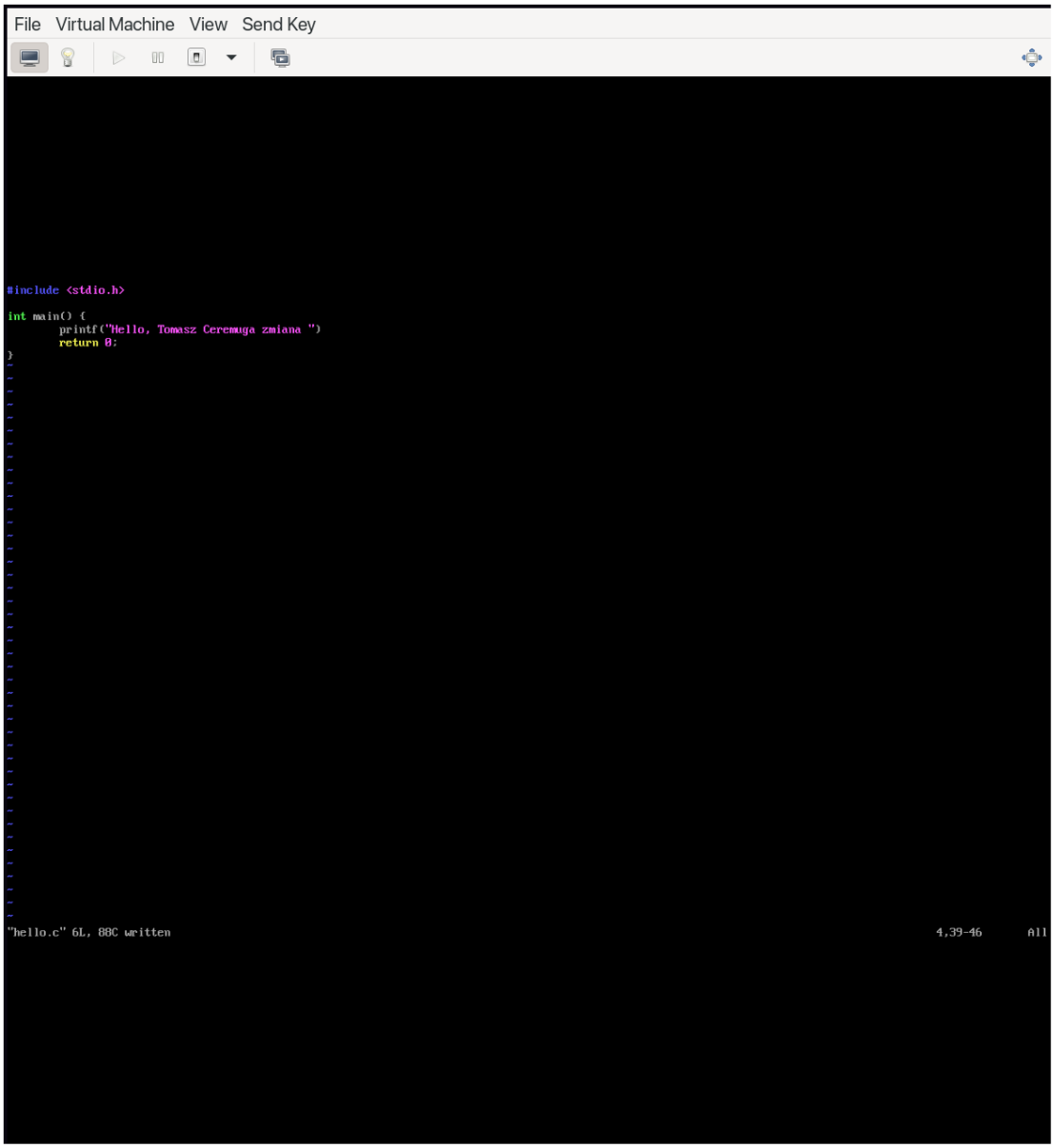
Kompresja polega na zastępowaniu podobnych wzorców, kiedy plik jest bardzo chaotyczny, jak np ten losowy, jest mało rzeczy które można skrócić

14.


File Virtual Machine View Send Key
[root@localhost lab2]# sudo tar -czvf logi_$(date +%F).tar.gz $(sudo find /var/log -type f -name "*.log" -mtime -7)
tar: Removing leading `/' from member names
/var/log/audit/audit.log
tar: Removing leading `/' from hard link targets
/var/log/tuned/tuned.log
/var/log/dnf.log
/var/log/dnf.librepo.log
/var/log/dnf.rpm.log
/var/log/anaconda/anaconda.log
/var/log/anaconda/X.log
/var/log/anaconda/program.log
/var/log/anaconda/packaging.log
/var/log/anaconda/storage.log
/var/log/anaconda/lvm.log
/var/log/anaconda/dnf.librepo.log
/var/log/anaconda/hawkey.log
/var/log/anaconda/dbus.log
/var/log/anaconda/ks-script-4bsrva_c.log
/var/log/anaconda/ks-script-96lpz1_4.log
/var/log/anaconda/ks-script-kh4oo5h0.log
/var/log/anaconda/ks-script-vgo7y8aa.log
/var/log/anaconda/journal.log
/var/log/boot.log
/var/log/kdump.log
/var/log/hawkey.log
[root@localhost lab2]#


File Virtual Machine View Send Key
[root@localhost lab2]# tar -xzvf logi_$(date +%F).tar.gz var/log/boot.log
var/log/boot.log
[root@localhost lab2]# _

15.


File Virtual Machine View Send Key
#include <stdio.h>
int main() {
printf("Hello, Tomasz Ceremuga zmiana ")
return 0;
}
"hello.c" 6L, 88C written
4,39-46 All


File Virtual Machine View Send Key
#include <stdio.h>
int main() {
printf("Hello, Tomasz Ceremuga cos ")
}
[root@localhost programy]# _

17.


File Virtual Machine View Send Key
[root@localhost ~]# cut -d: -f7 /etc/passwd | sort | uniq -c | sort -rn | head -3 > ./lab2/powloki.191337
[root@localhost ~]# cat .lab2/powloki.191337
cat: .lab2/powloki.191337: No such file or directory
[root@localhost ~]# cat ./lab2/powloki.191337
25 /sbin/nologin
1 /sbin/shutdown
1 /sbin/halt
[root@localhost ~]#